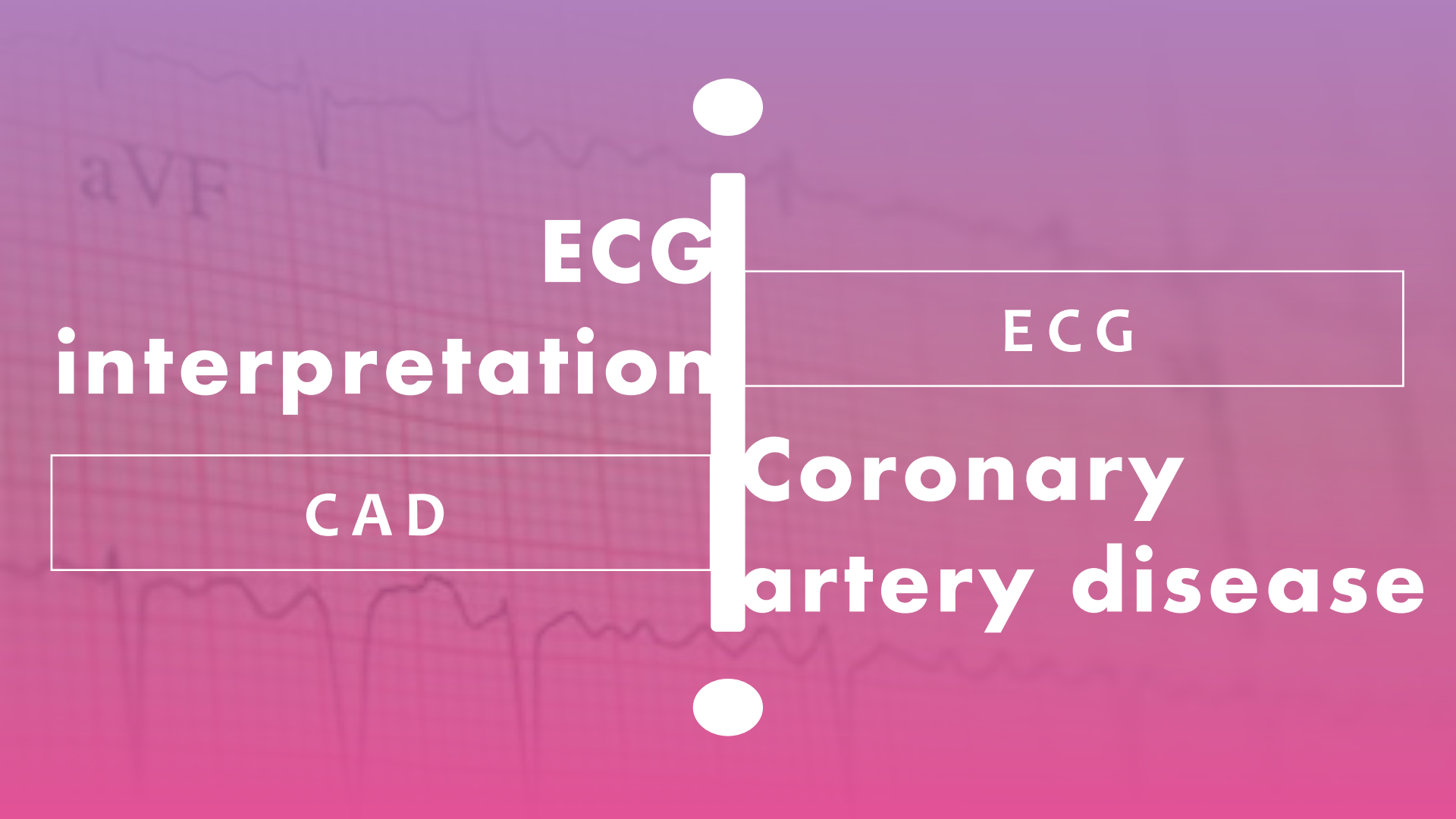


ECG in clinical practice

Coronary artery disease

Treechada Wisaratapong, M.D.
Cardiac Electrophysiologist
Songklanagarind hospital
Prince of Songkla University



ECG
interpretation

ECG

CAD

**Coronary
artery disease**

ECG

interpretation

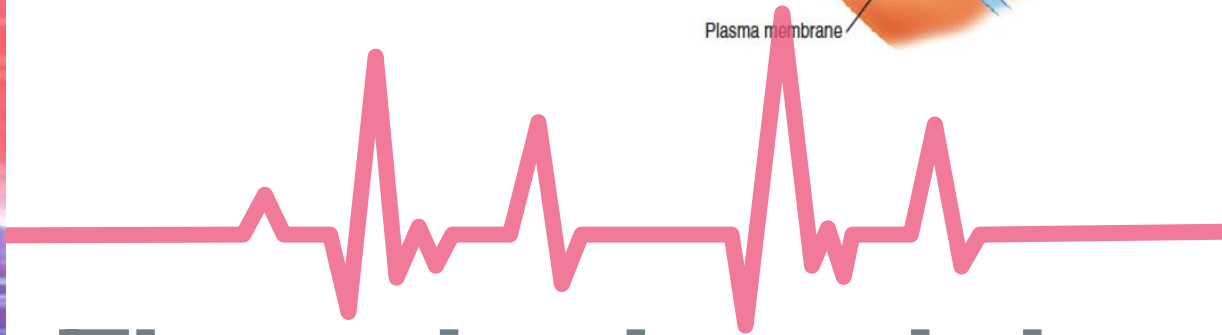
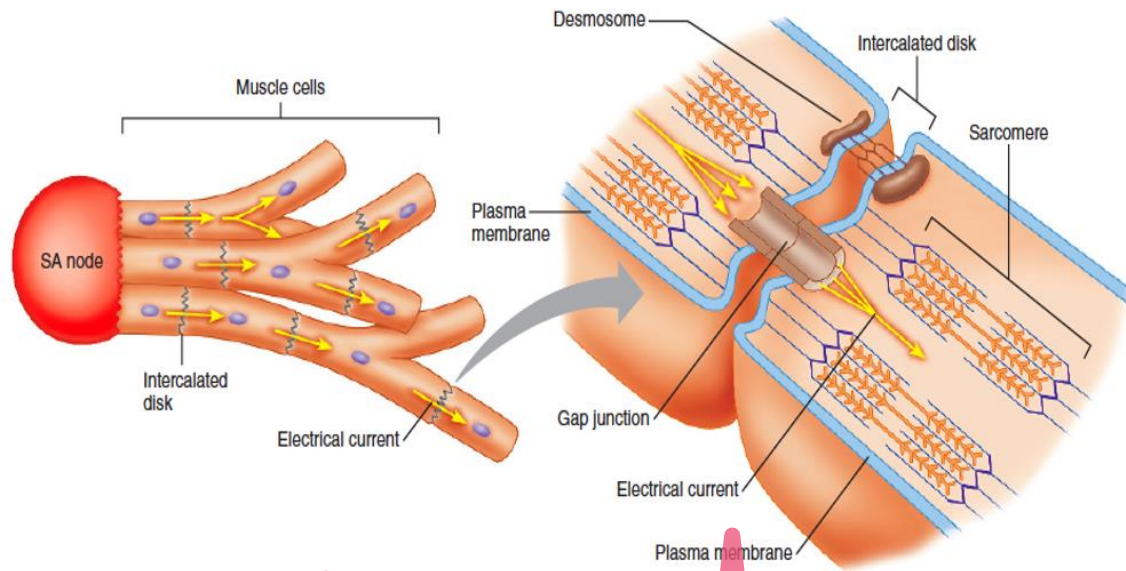
- Action potential
- Vector
- ECG waves & intervals
- Surface ECG leads
- Einthoven's triangle
- Special leads

CAD

ECG

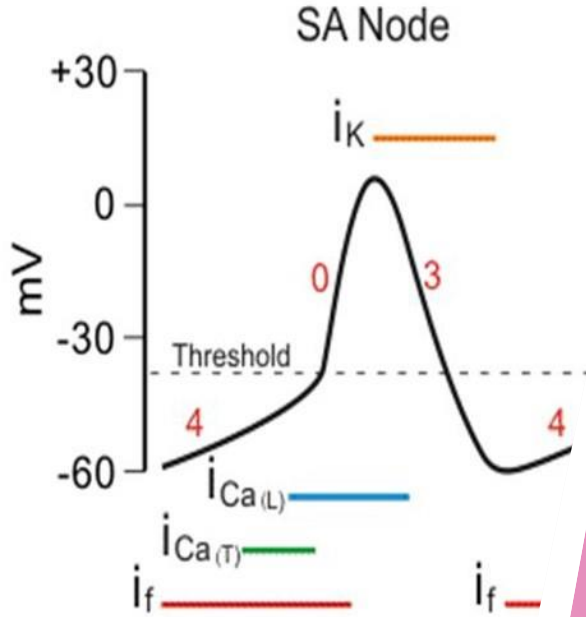
- Spectrum of CAD
- Pathophysiology
- Pattern of ischemic injury
- Clinical presentation & ECG correlation

Coronary artery disease

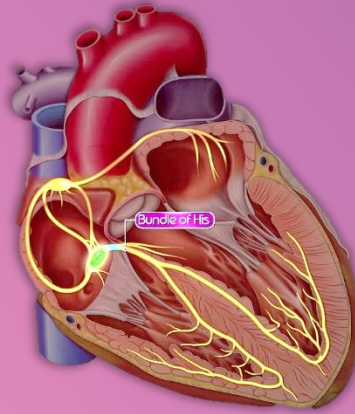


Electrical activity

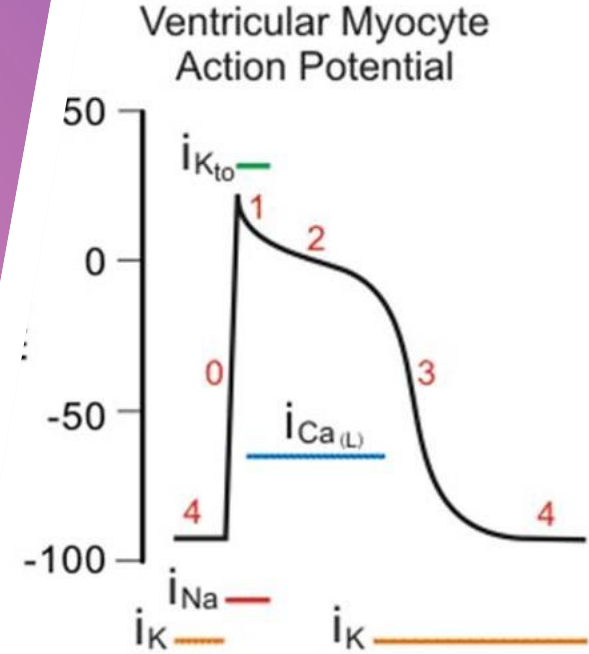
Action potential



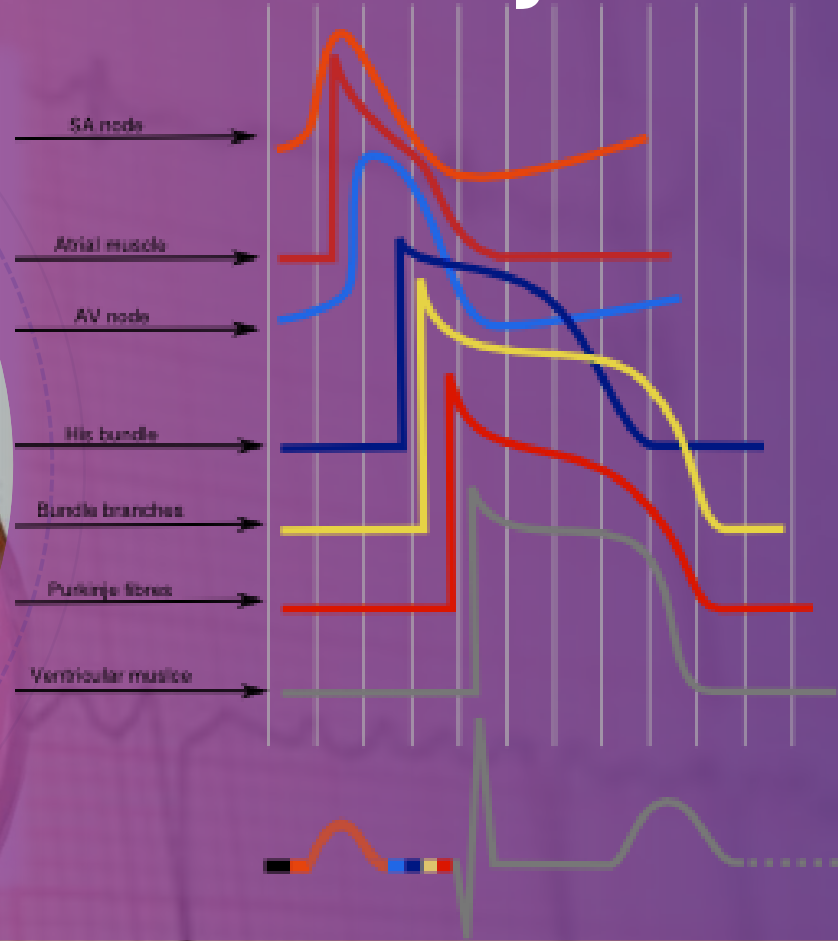
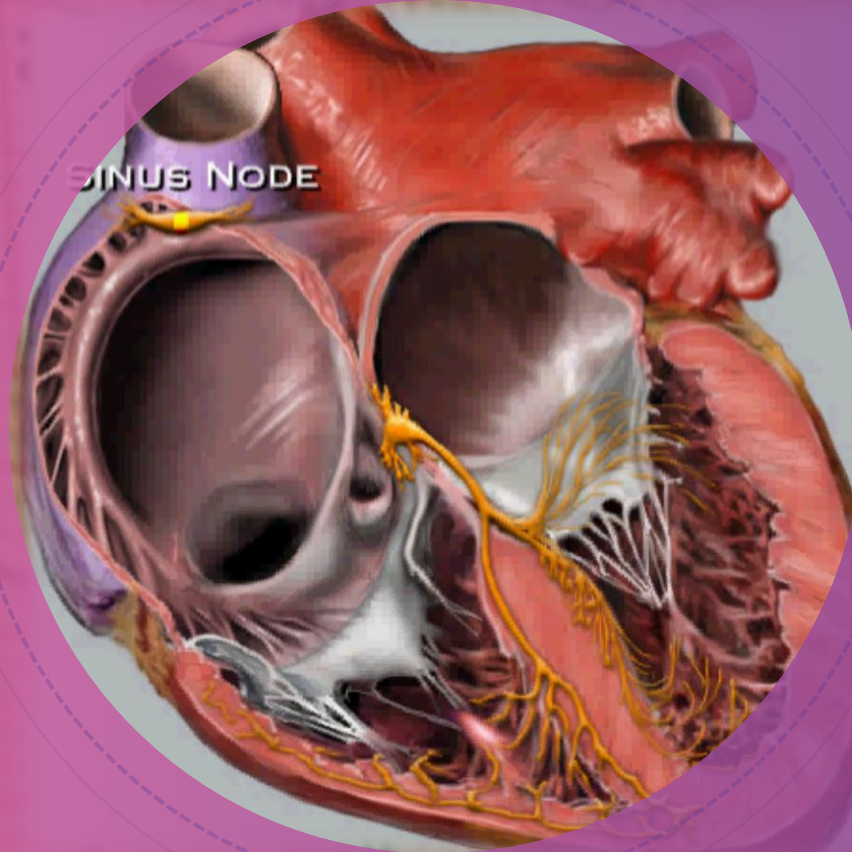
Pacemaker cells:
SA node, AV node



Cardiac muscle cells:
Atrium, His bundle,
purkinje, ventricle

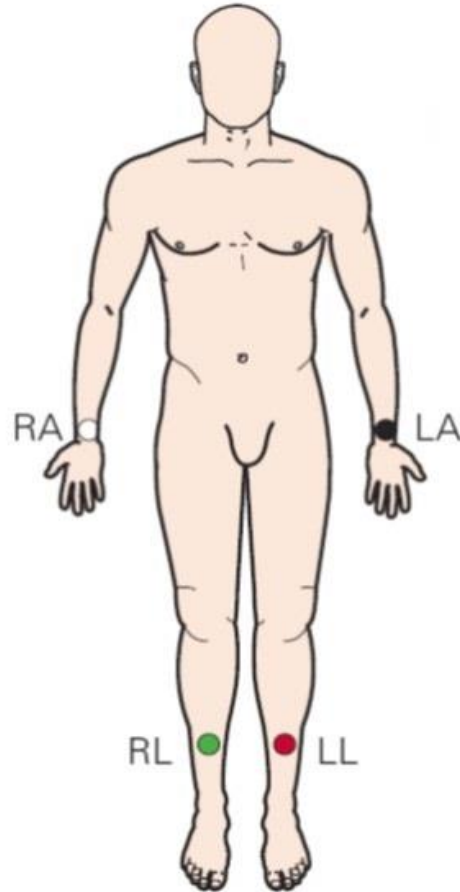


Cardiac conduction system

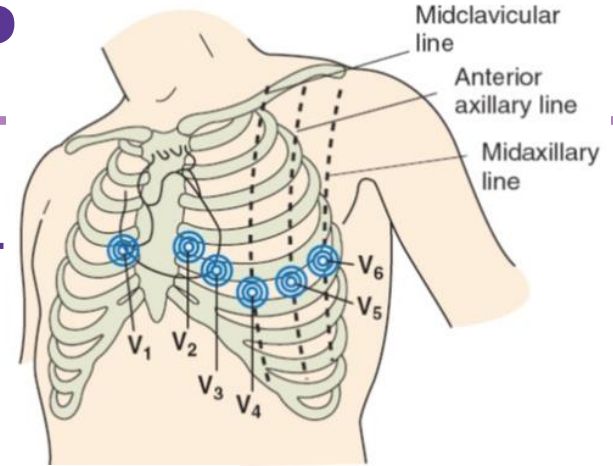


Surface ECG leads

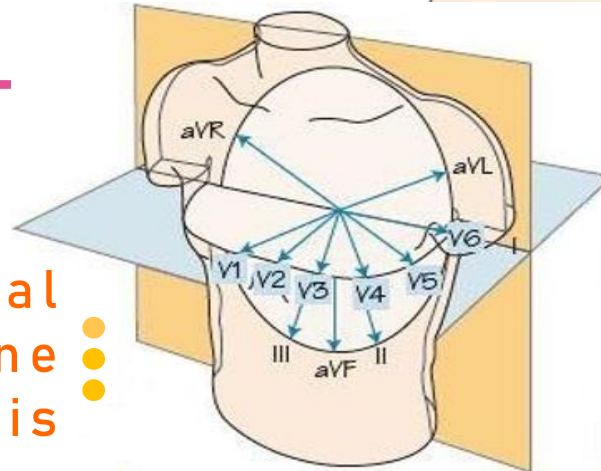
Limb leads



Precordial chest leads

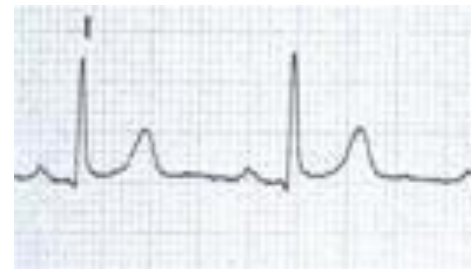
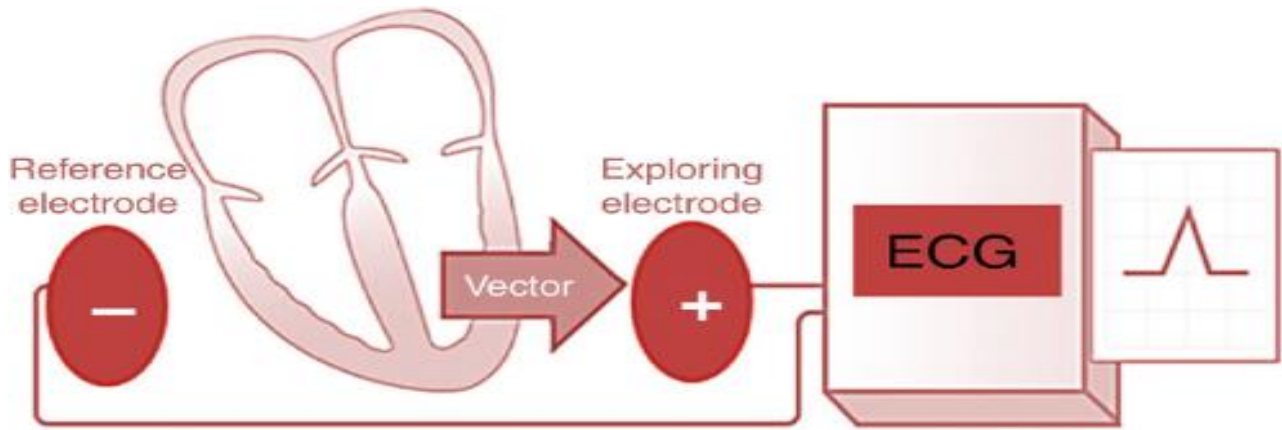


Frontal plane axis



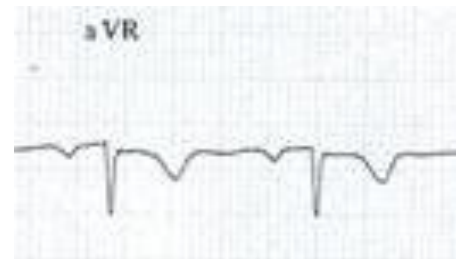
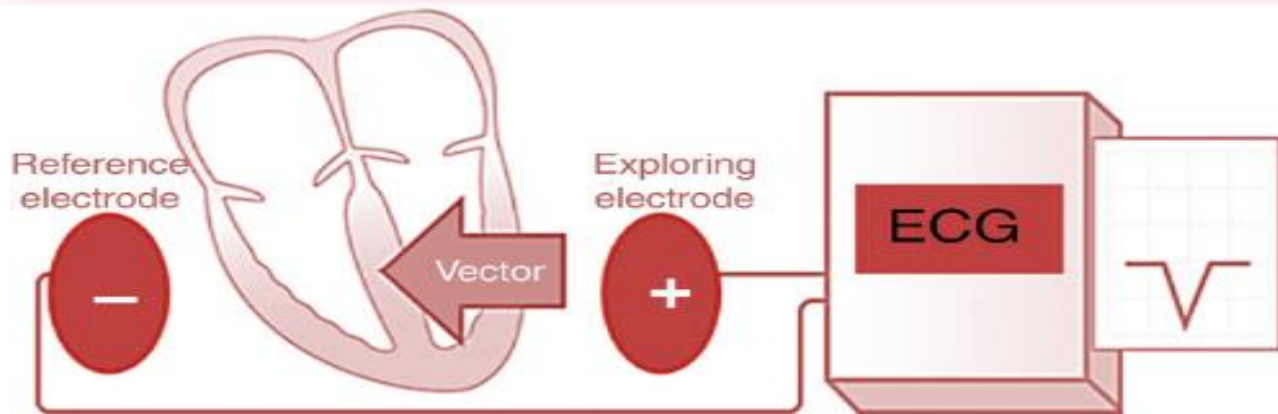
Horizontal plane axis





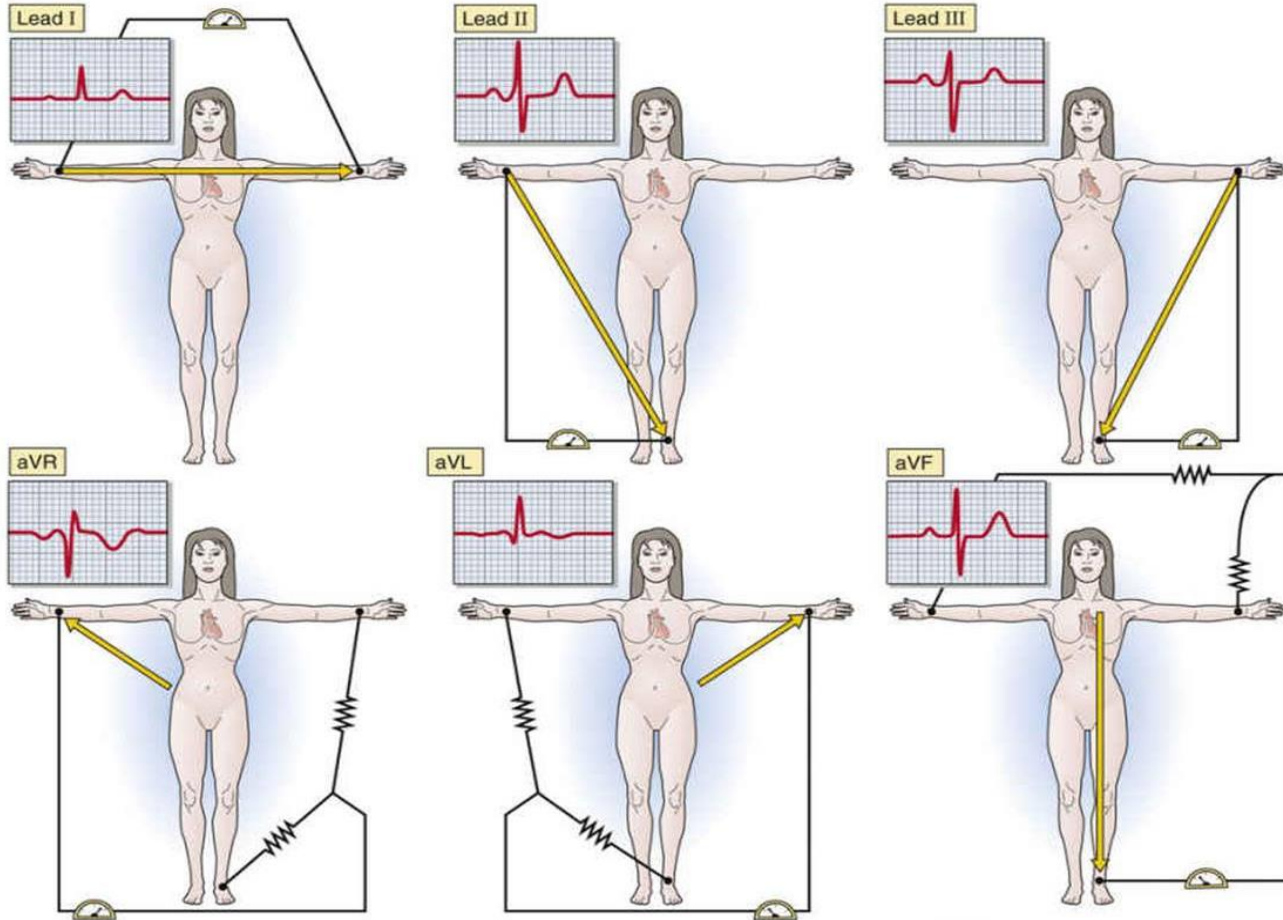
Positive wave

Vector & ECG leads

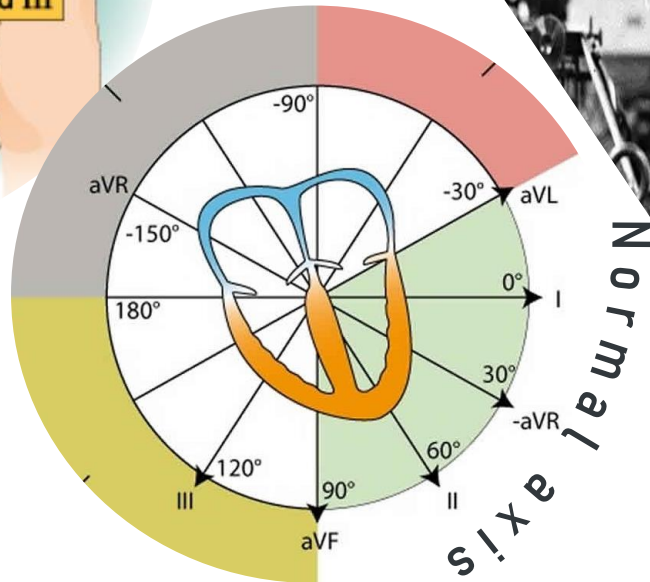
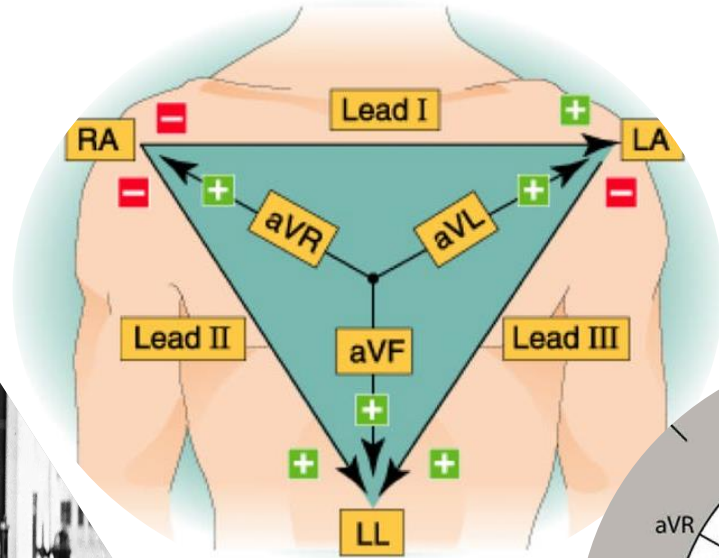


Negative wave

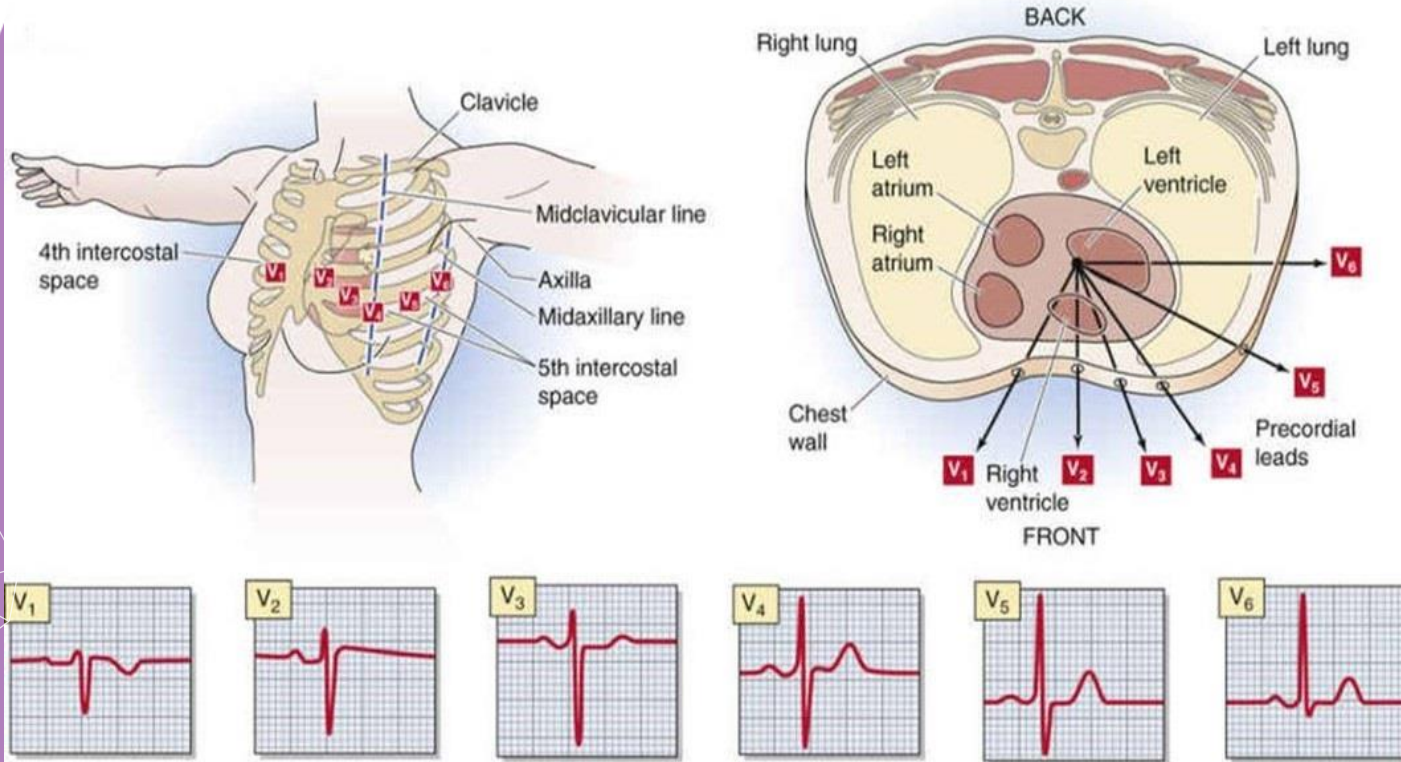
Bipolar & Unipolar Limb Leads



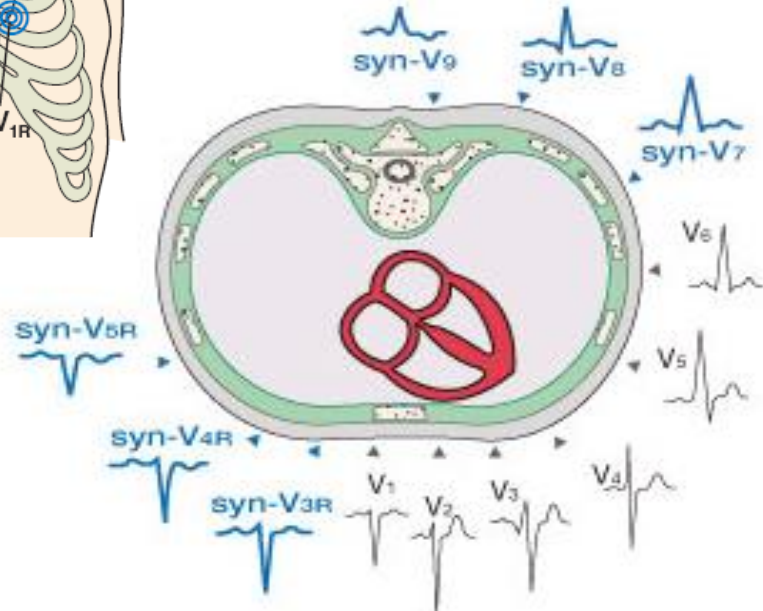
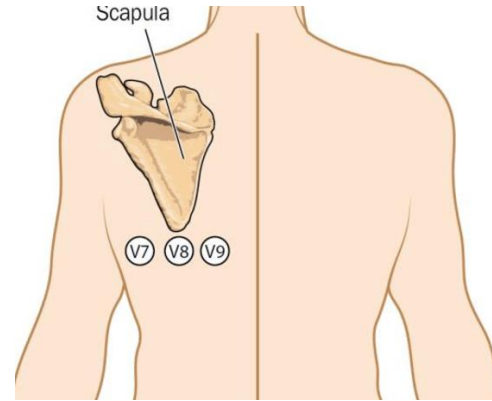
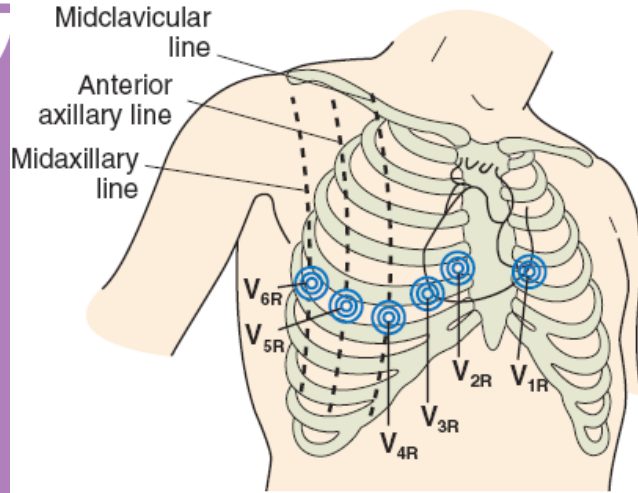
Einthoven's Triangle



Unipolar Precordial Chest Leads



Special leads



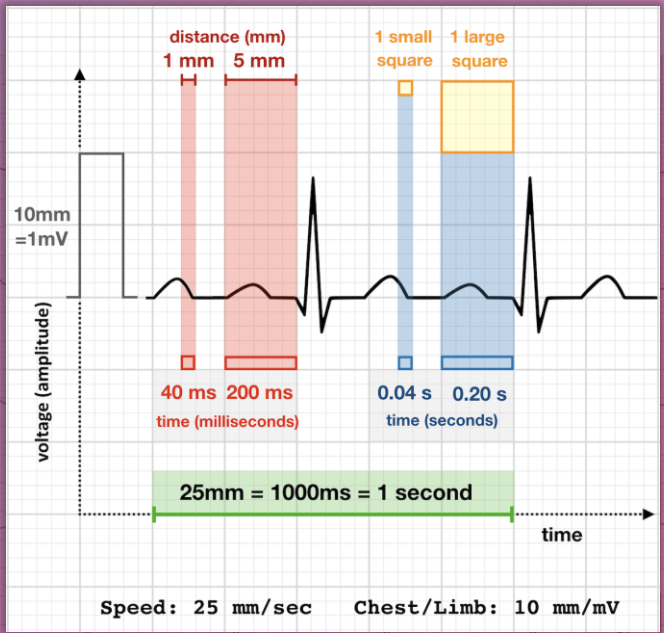
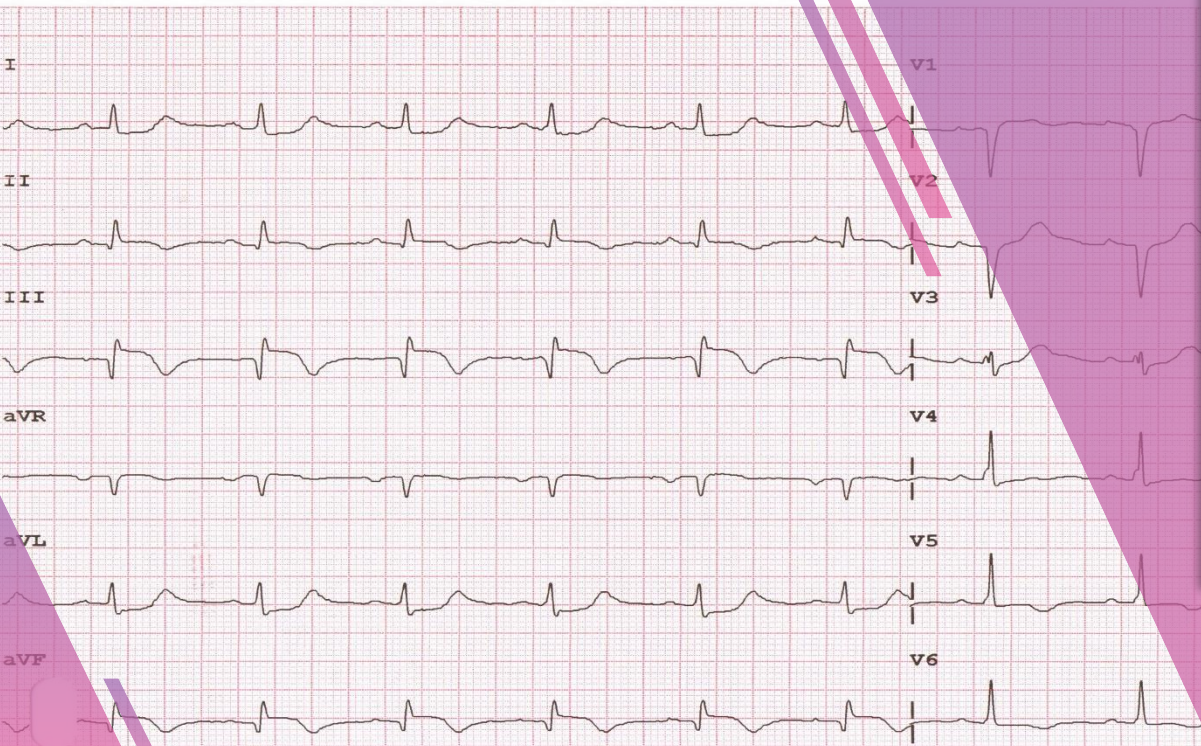
Amplitude calibration

Rate 74
PR 172
QRSD 100
QT 436
QTc 484

AGE IS NOT ENTERED, ASSUMED TO BE 1 YEARS OLD FOR PURPOSE OF ECG INTERPRETATION
SINUS BRADYCARDIA
FIRST DEGREE AV BLOCK
REPOLARIZATION ABNORMALITY SUGGESTS LVH
PROLONGED QT, PROBABLY SECONDARY TO WIDE QRS

--AXIS--
P 18
QRS 44
T -57

- ABNORMAL ECG -



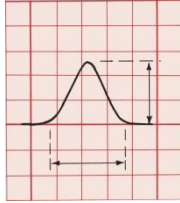
Paper speed

Dev: Speed: 25 mm/sec Limb: 10 mm/mV Chest: 10 mm/mV

F 50~ 0.5- 40 Hz W H090A

P ATRIAL
DEPOLARIZATION

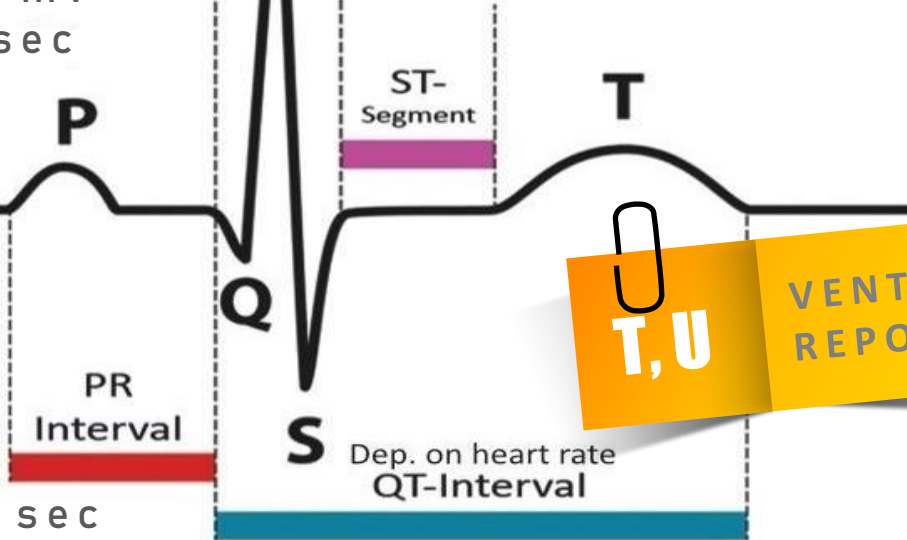
Amplitude < 0.25 mV
Duration < 0.12 sec



QRS COMPLEX

QRS VENTRICULAR
DEPOLARIZATION

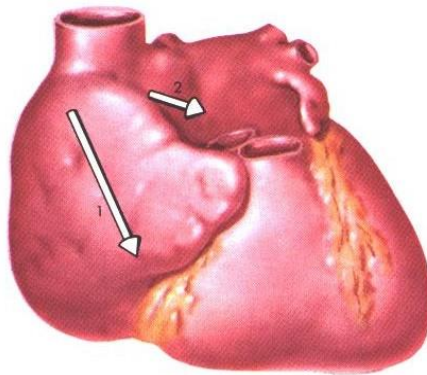
QRS interval < 0.12 sec



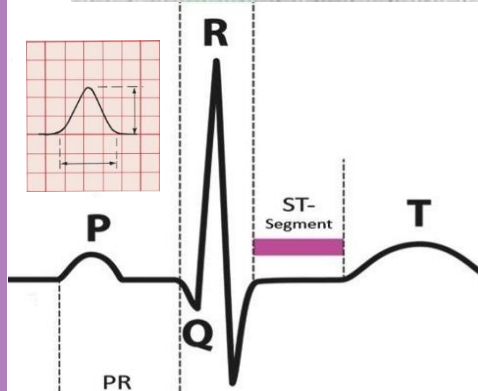
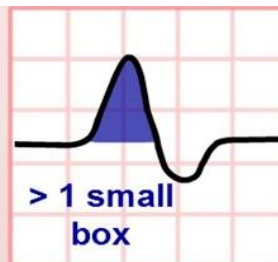
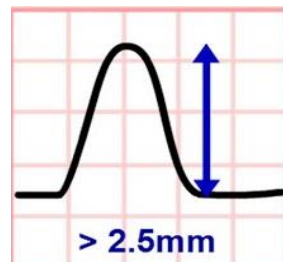
PR interval < 0.20 sec

T,U VENTRICULAR
REPOLARIZATION

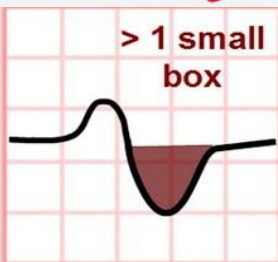
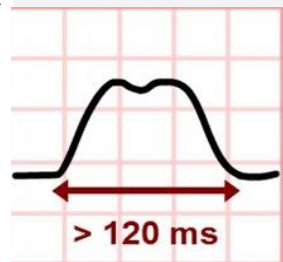
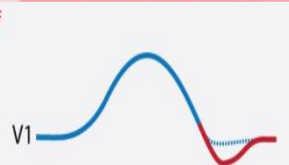
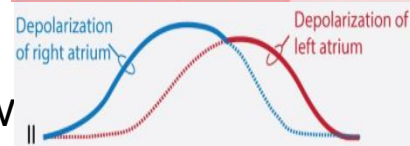
Atrial depolarization = P wave



Tall and peak P wave in lead II, III, aVF > 0.25 mV

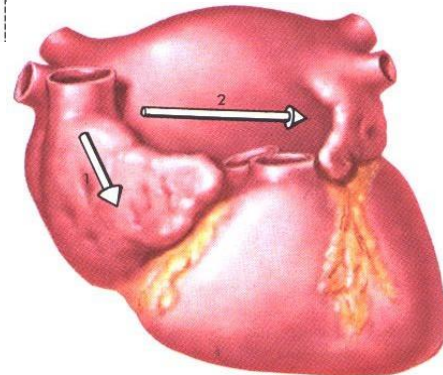
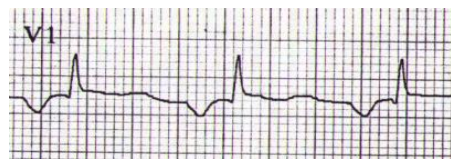
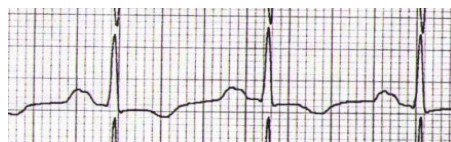


Amplitude ≤ 0.25 mV
Duration ≤ 0.12 sec



Broad and notch P wave > 0.12 sec in lead II

Negative terminal portion of P wave in V1 ≥ 1 mm²

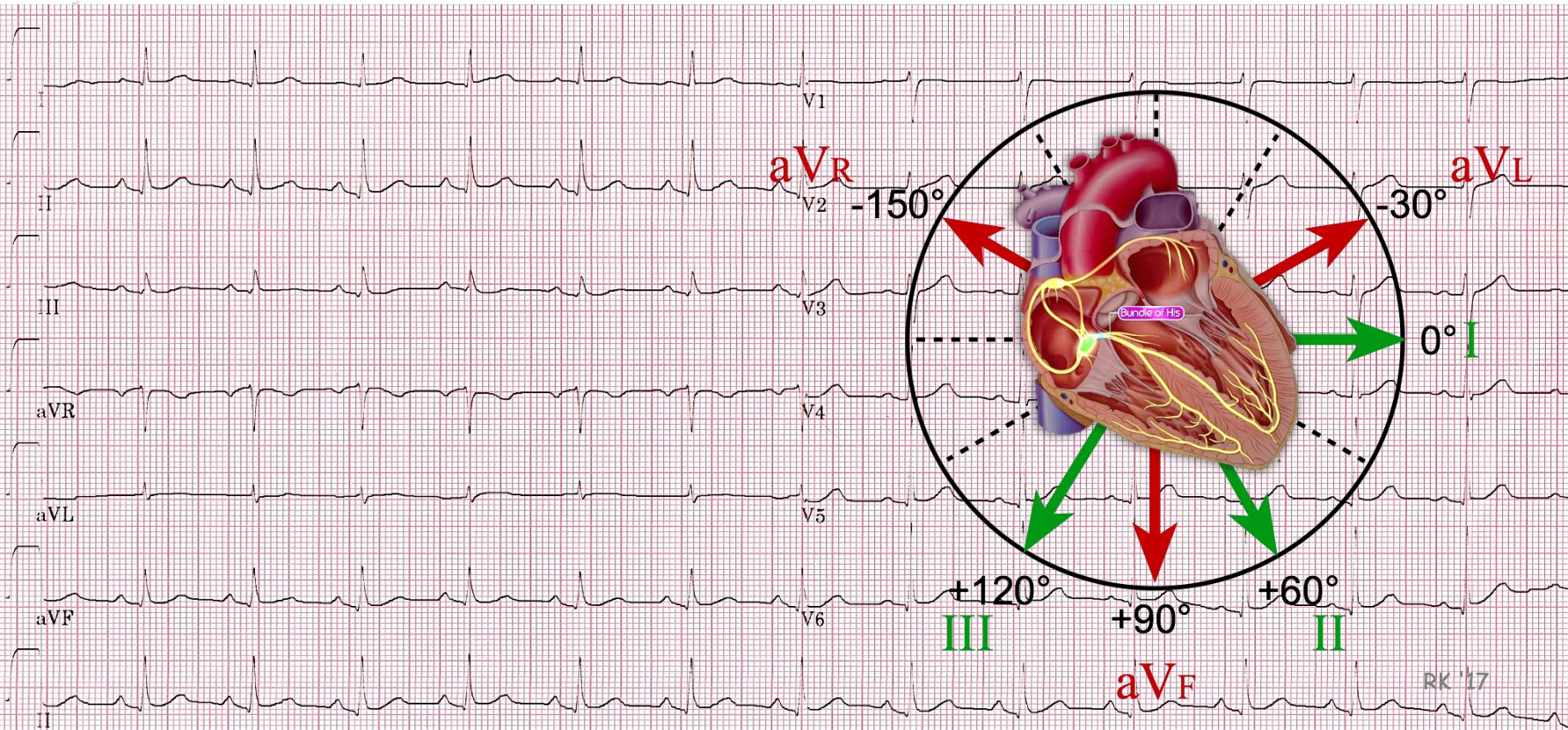


1-Jan-1955
Male

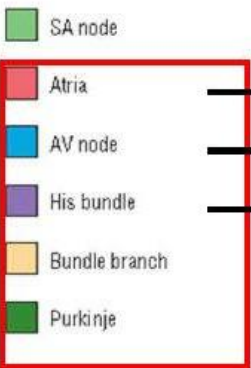
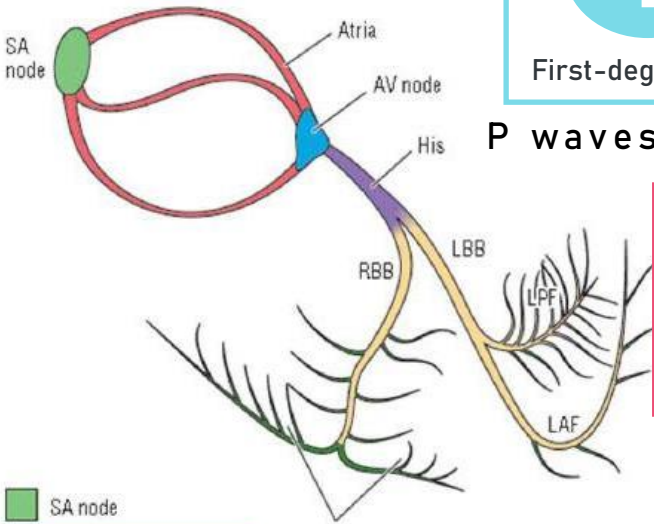
Vent. rate 83 bpm
PR interval 172 ms
QRS duration 80 ms
QT/QTc 360/423 ms
P-R-T axes 71 54 58

Normal sinus rhythm
Normal ECG

Normal P wave from SA node



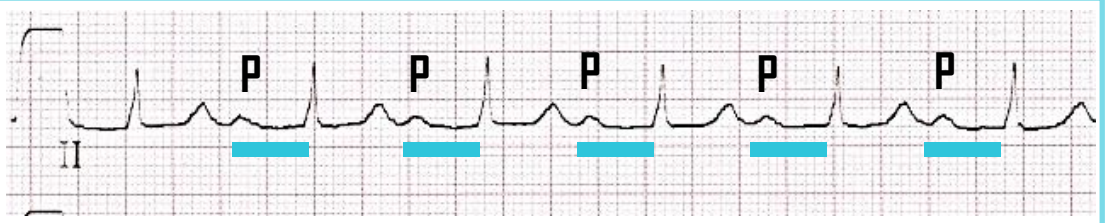
PR interval
 $\leq 0.2 \text{ sec}$



→ Atrial conduction delay
 → AV nodal conduction delay
 → Intra-Hisian conduction delay
 }
 } Infra-Hisian conduction delay
 }
 } Infra-nodal conduction delay

1

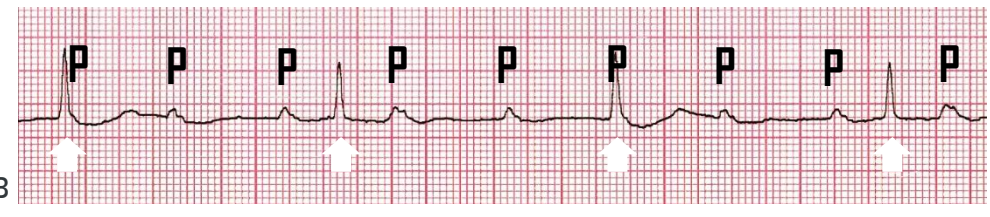
First-degree AVB



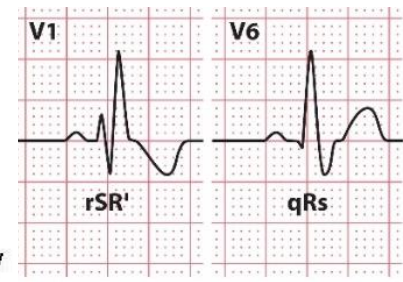
P waves associated with 1:1 AV conduction, PR interval > 200 ms

3

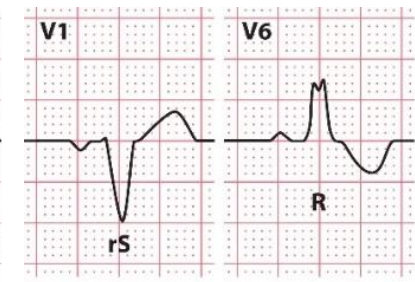
Third-degree AVB



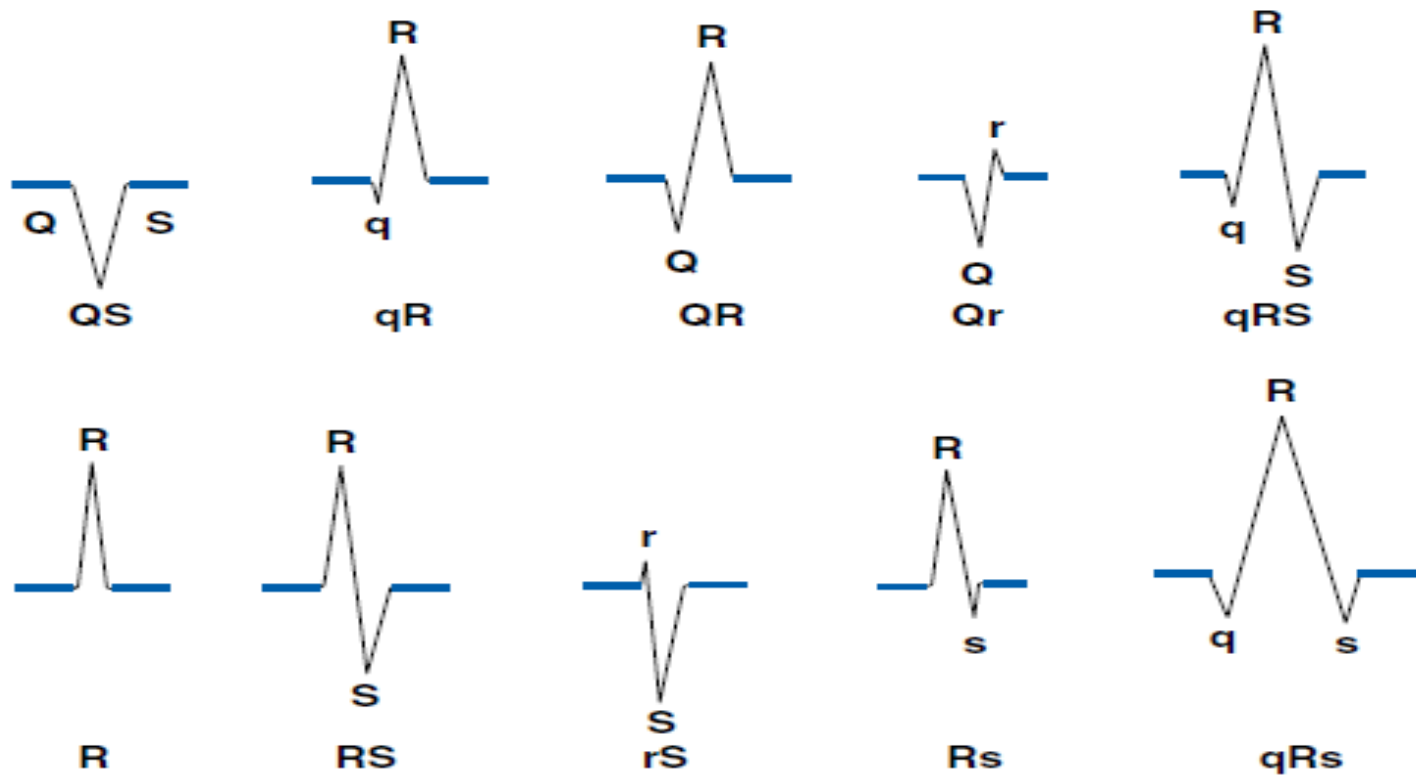
P wave and QRS dissociation



RIGHT BUNDLE BRANCH BLOCK



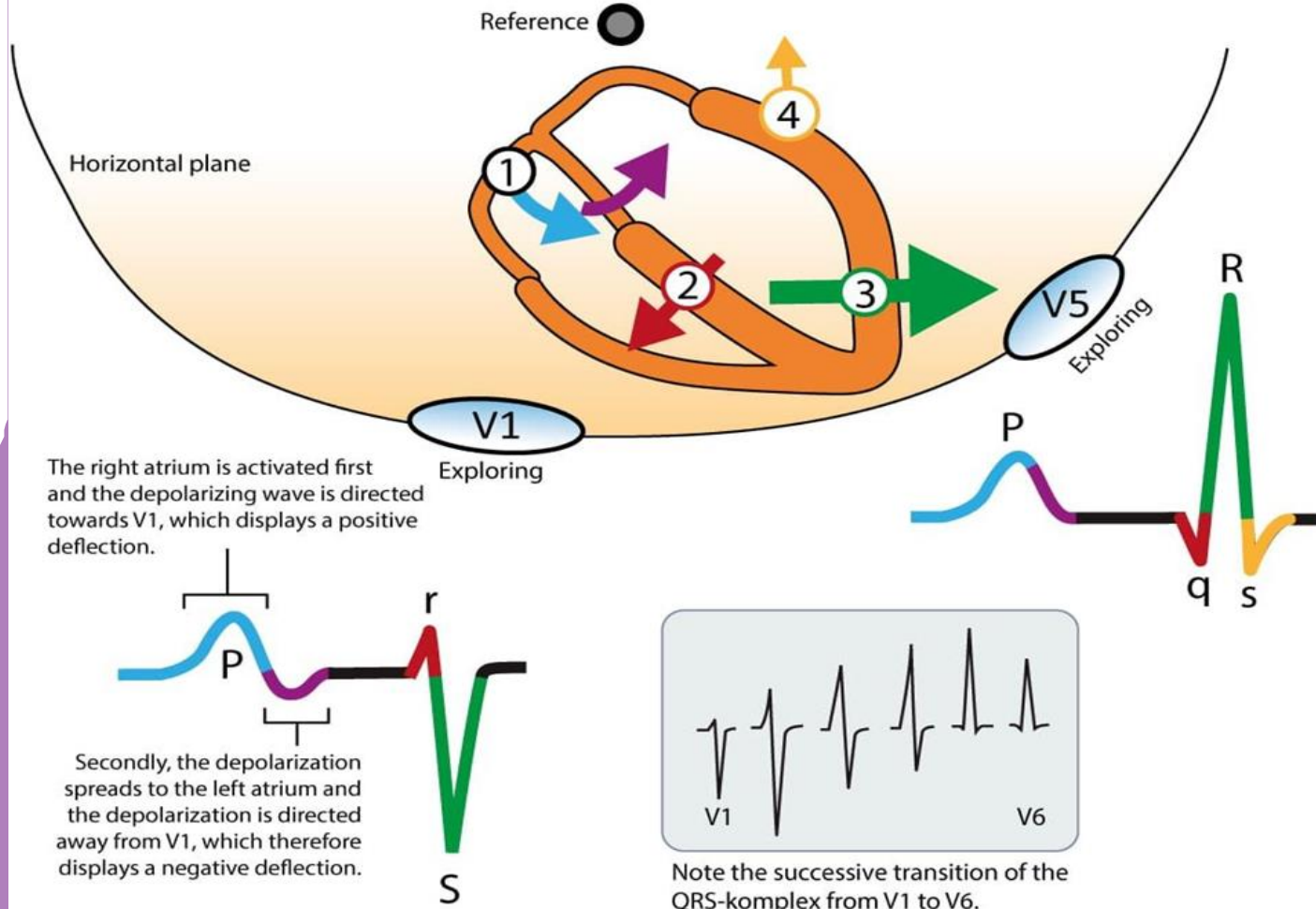
LEFT BUNDLE BRANCH BLOCK

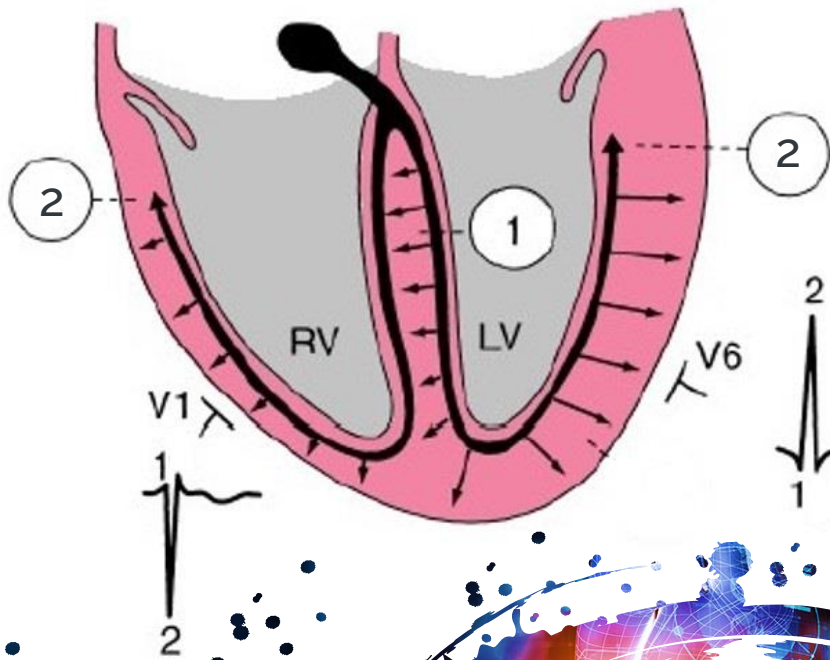


QRS complex nomenclature

Q wave: 1st negative deflection, R wave: 1st positive deflection, S wave: 1st negative deflection after R wave, R' wave: 2nd positive wave after S wave

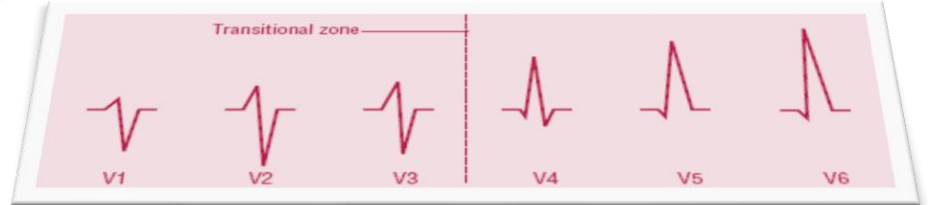
Precordial QRS formation





01

Septal depolarization (left to right)



02

LV lateral wall and RV free wall depolarization (right to left)

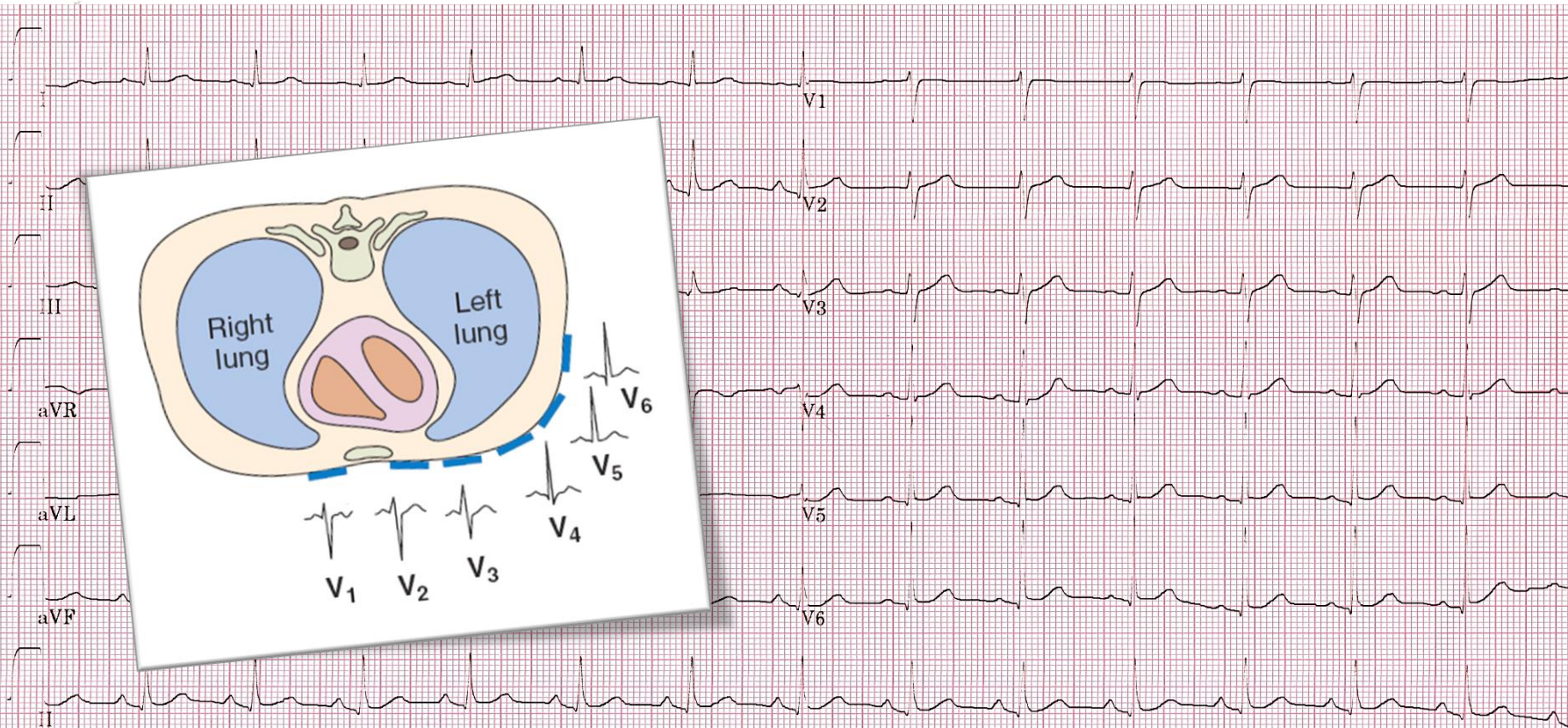
Precordial QRS formation

1-Jan-1955
Male

Vent. rate 83 bpm
PR interval 172 ms
QRS duration 80 ms
QT/QTc 360/423 ms
P-R-T axes 71 54 58

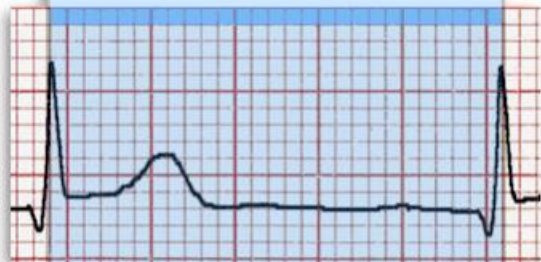
Normal sinus rhythm
Normal ECG

12-lead ECG





27 **SMALL** squares



$$\frac{1500}{27} = 55.6$$

= 56 bpm

5.4 **LARGE** squares



$$\frac{300}{5.4} = 55.6$$

= 56 bpm

9 x **R waves**

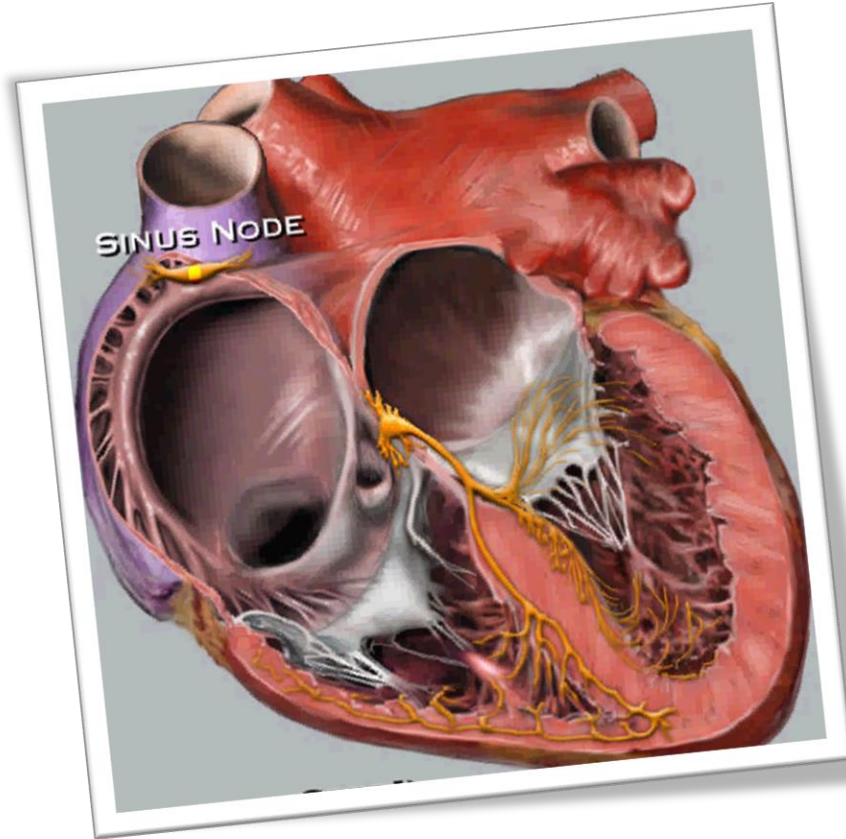


9 beats in 10 seconds
= 9 x 6 beats in 1 minute

= 54 bpm

Speed: 25 mm/sec

Pacemaker cell/Automaticity



SA node : 60-100 bpm

Internodal pathways : between
SA and AV node

AV node : 40-60 bpm

Bundle of His : below AV node

Left bundle branch (LBB)

Right bundle branch (RBB)

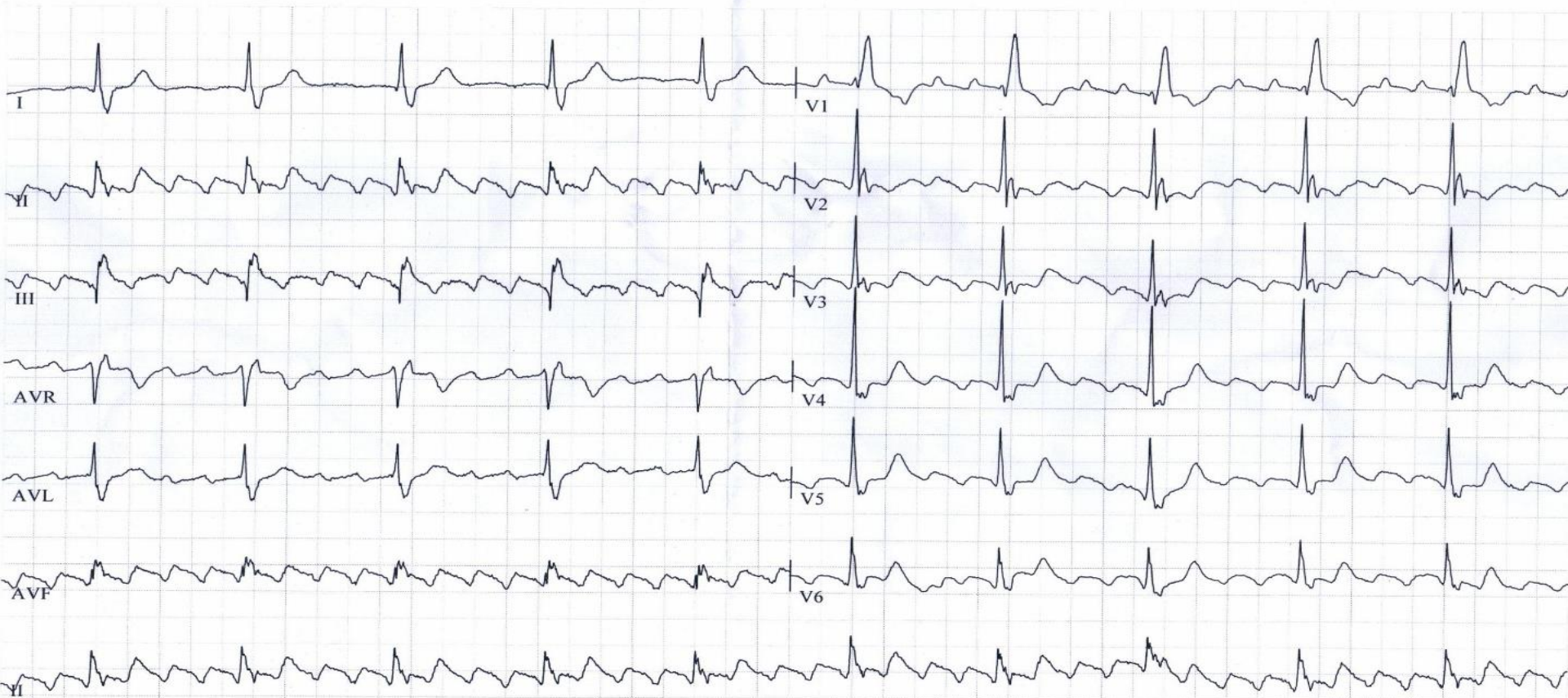
Purkinje system : network of
fibers, 20-40 bpm

Rhythm

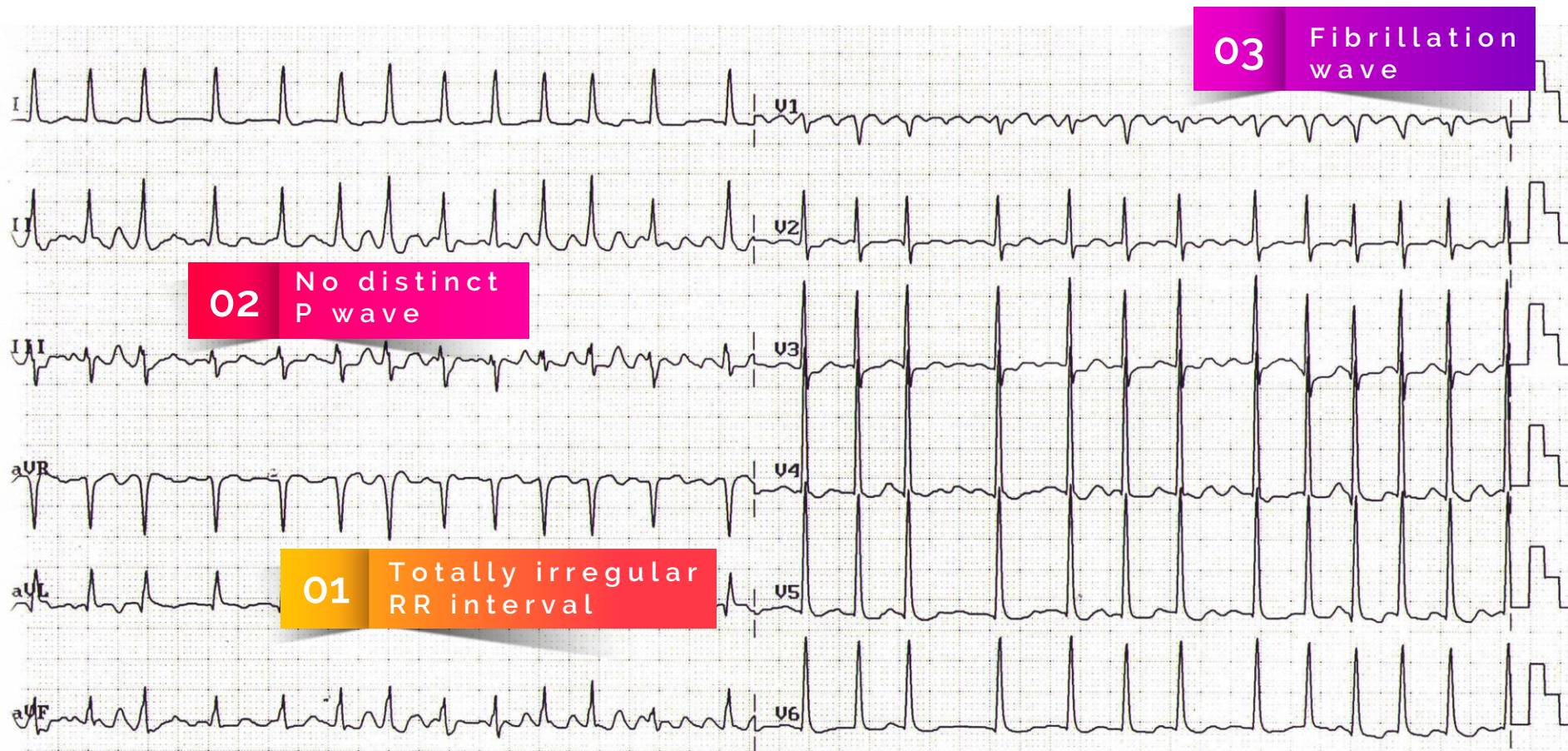
- **Sinus rhythm** : normal sinus rhythm, sinus bradycardia, sinus tachycardia
- **Atrial rhythm** : atrial fibrillation, atrial flutter, atrial tachycardia
- **Junctional rhythm**
- **Ventricular rhythm** : ventricular tachycardia, ventricular fibrillation
- **Paced rhythm**



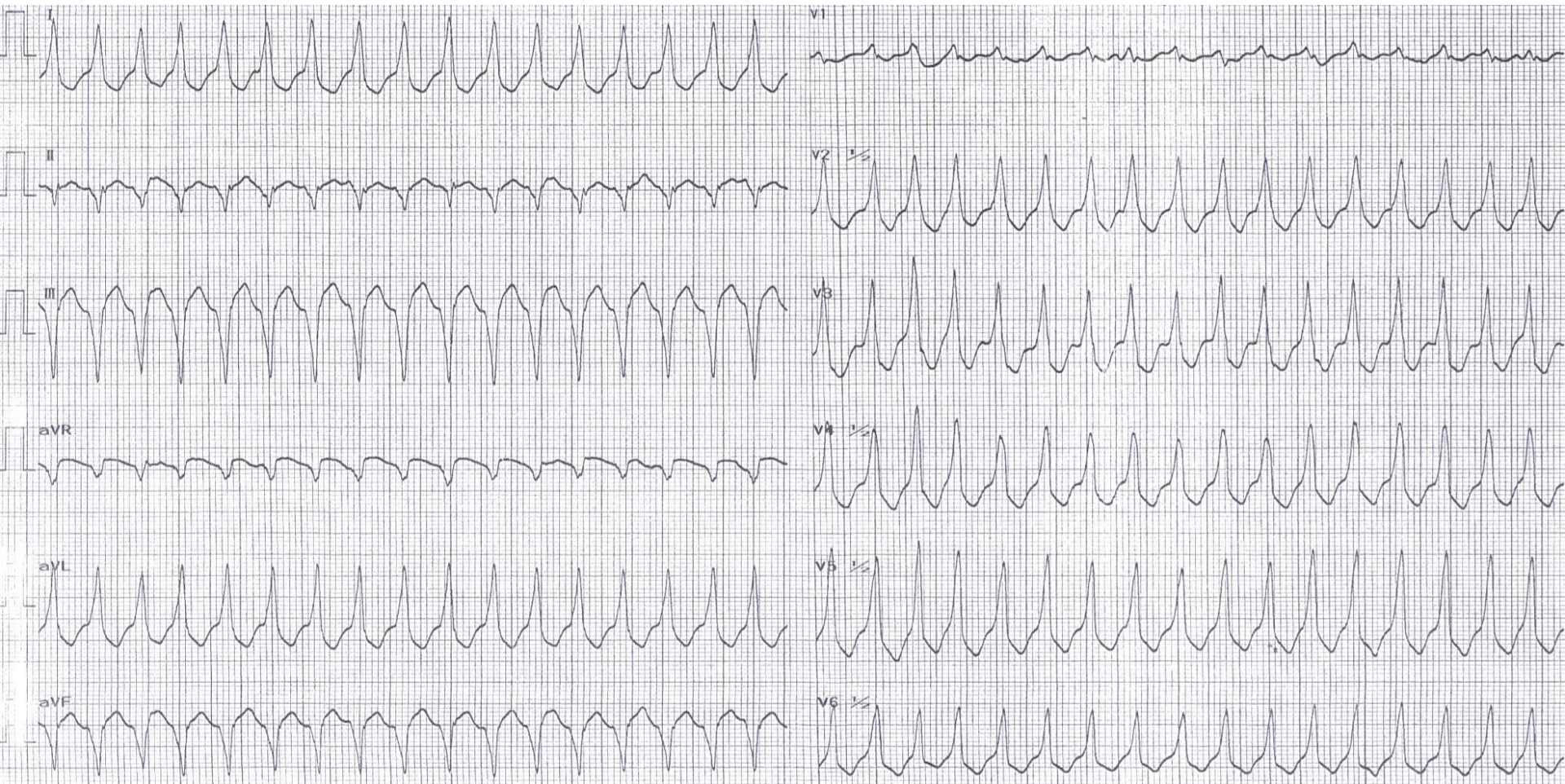
ATRIAL FLUTTER



ATRIAL FIBRILLATION



VENTRICULAR TACHYCARDIA





Evaluation data

QRS (ms)	156
P (ms)	
PQ (ms)	
QT (ms)	509
QTc B. (ms)	542 (138%)
QTc disp. (ms)	32
QRS axis	-72?
HR	68
AVG RR (ms)	882

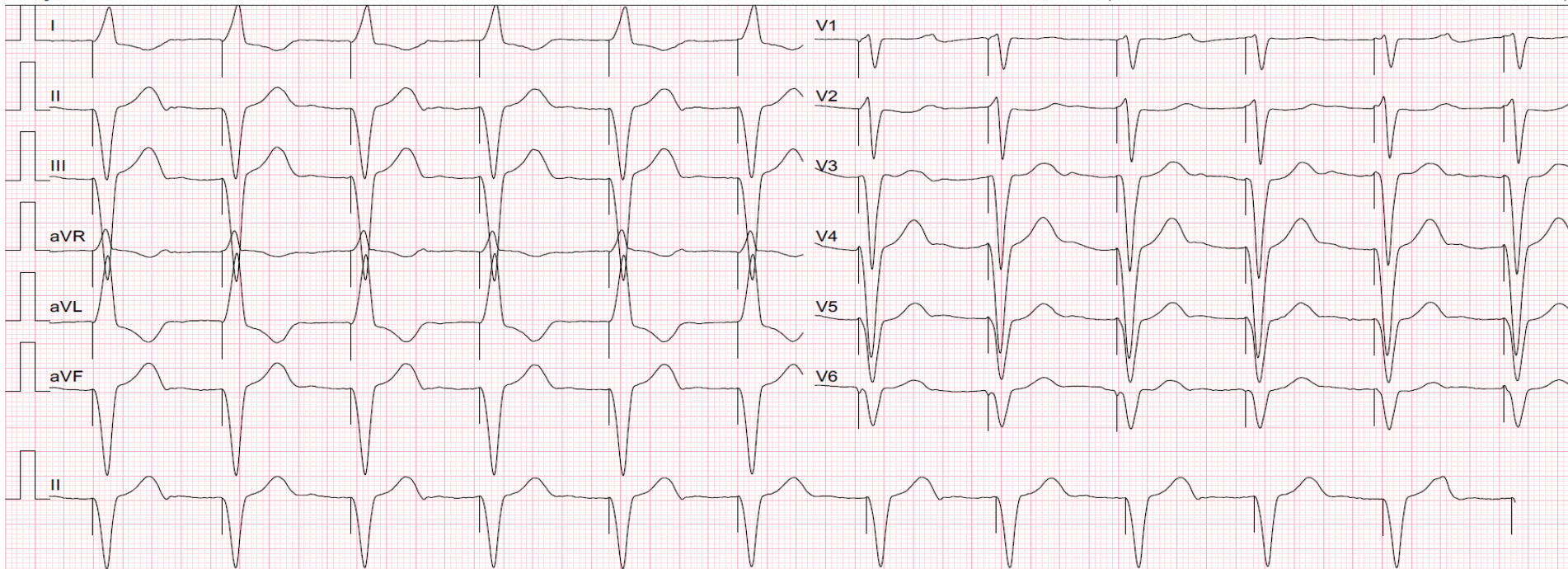
Unconfirmed Report

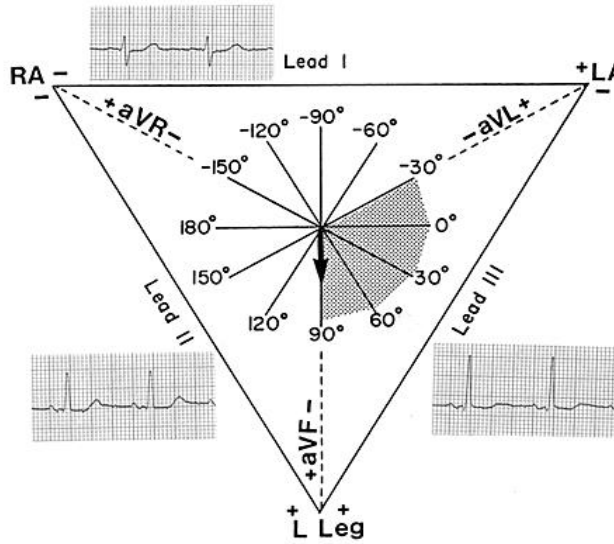
pacemaker ECG

Paced rhythm

Analysed ECG HR: 68

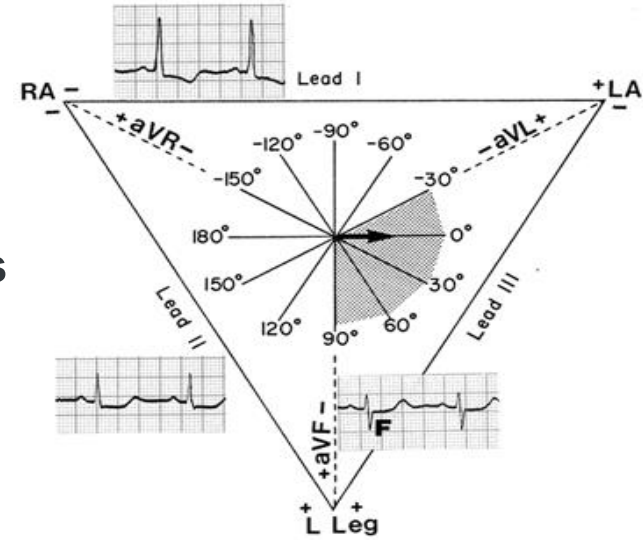
(25 mm/s 10 mm/mV 0.05Hz - 45Hz / 50Hz true wave®)





1. หา leads ที่มี vector biphasic

2. Axis จะตั้งฉากกับ leads ที่ biphasic คือถ้า lead ที่ตั้งฉากชี้ทางไหน axis คือทางนั้น



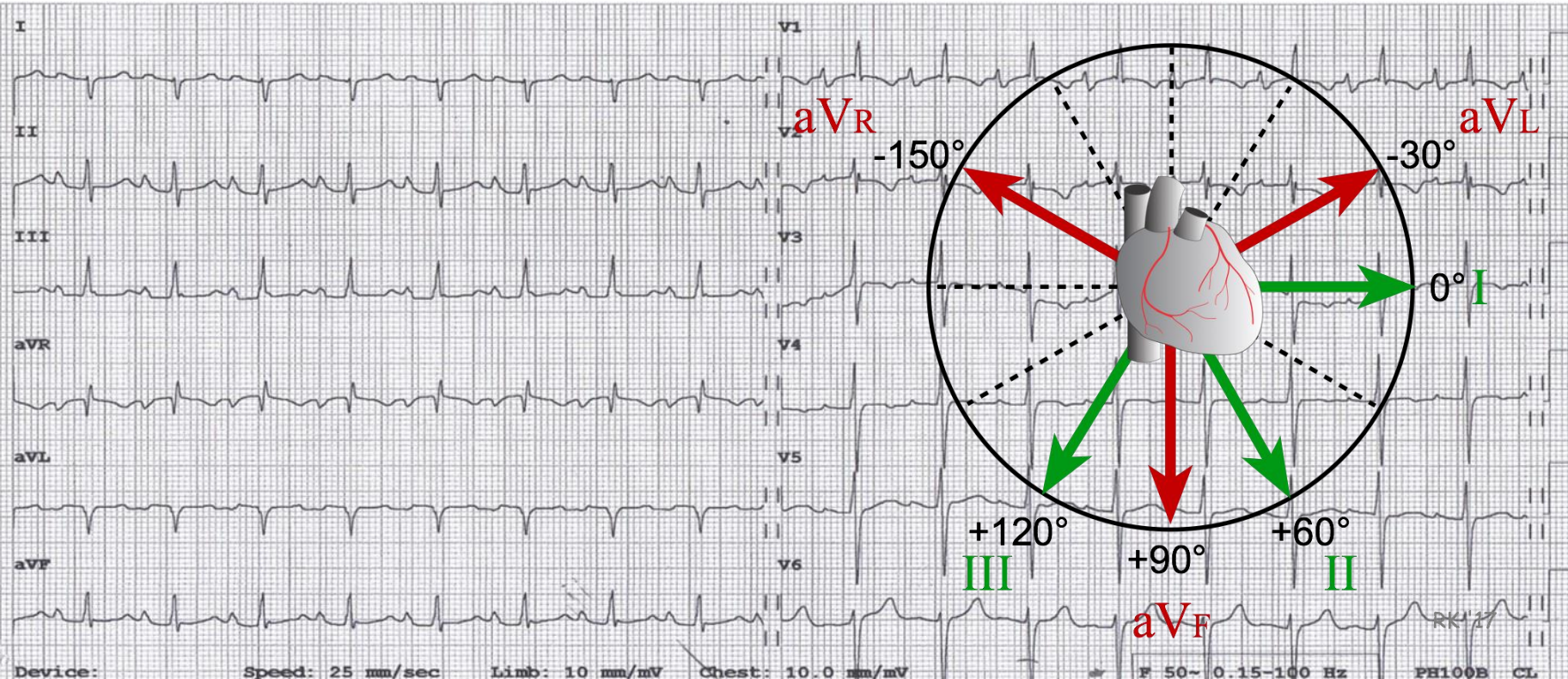
Frontal plane axis

Rate	105	. Age not entered assumed to be 50 years old for purpose of ECG interpretation
PR	199	. Sinus tachycardia.....rate> 99
QRSD	95	. Borderline prolonged PR interval.....PR >197, V-rate 91-120
QT	334	. LAE, consider biatrial enlargement.....P>80ms <-.15mV V1&.25mV limb lds
QTc	442	. RVH with secondary repolarization abnrm.....prom R/R', RAD/RAE & repol abnrm

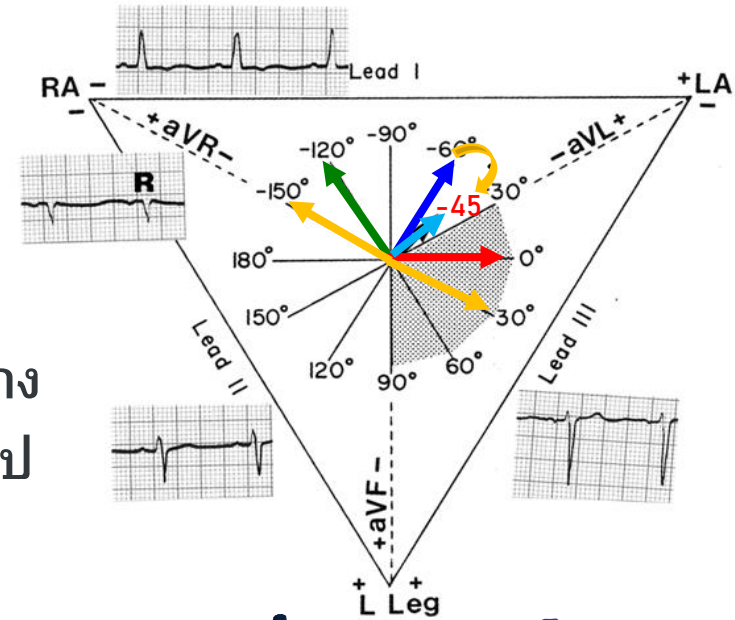
--AXIS--

P	25	Frontal plane axis = +120	- ABNORMAL ECG -
QRS	113		
T	50		
12 Lead; Standard Placement			

Unconfirmed Diagnosis



1. จำมุมของ Einthoven's triangle ให้ได้ก่อน รวมถึงคู่ตั้งฉาก
2. หา vector ตรงกลางของ lead I, II, III
3. ดู vector ที่ตั้งฉากกับ vector กลางว่ามีทิศทางไปด้านไหน จะตั้ง axis ของ vector กลางให้ไปด้านนั้นอีก 15 องศา

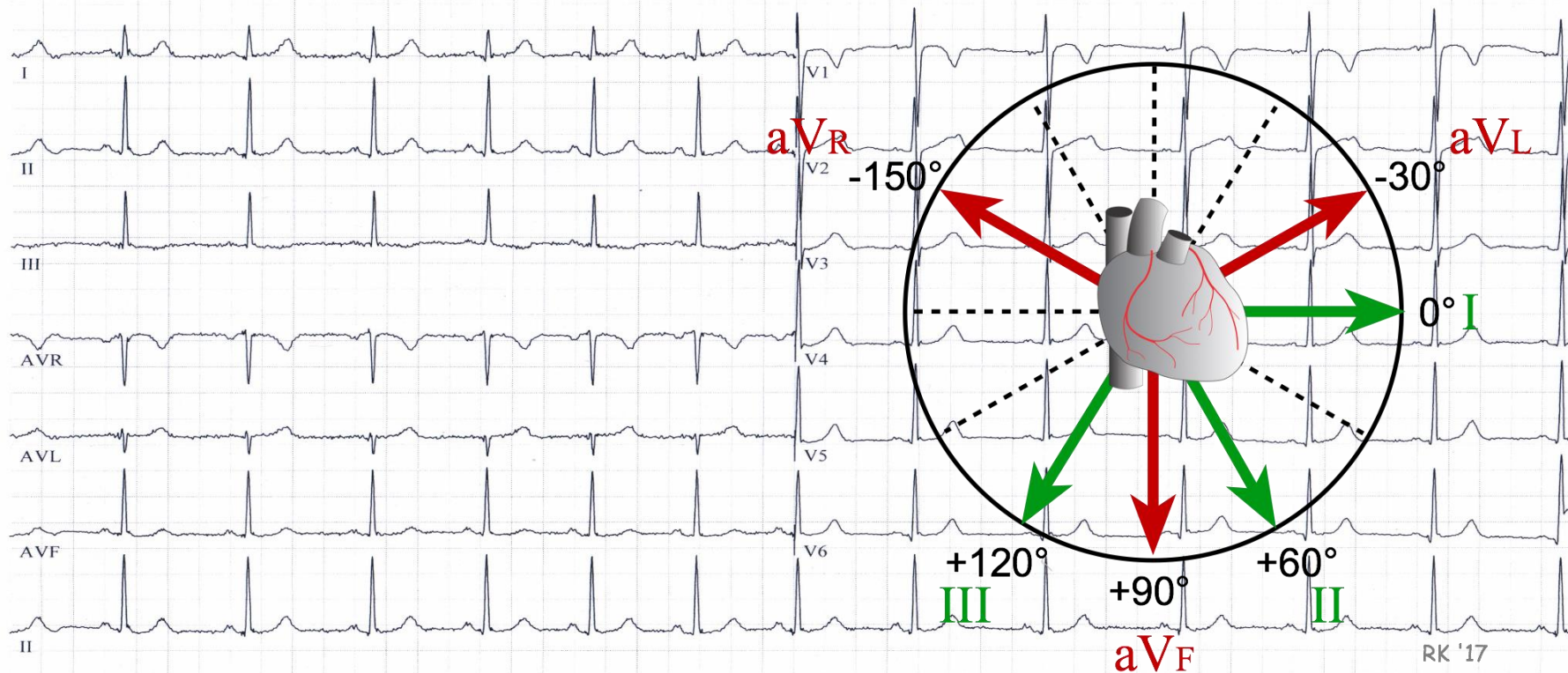


Frontal plane axis

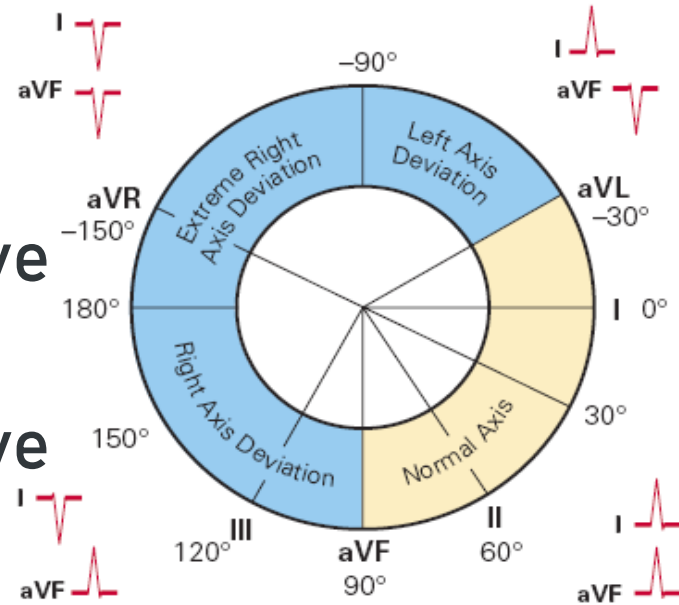
Female
27yrs
Vent. Rate 78bpm
PR interval 114ms
QRS duration 76ms
QT/QTc 380/433ms
P-R-T axes 43/72/22°
P duration 78ms
RR/PP interval 766/765ms

System Evaluation:
Normal sinus rhythm with sinus arrhythmia
Normal ECG

Frontal plane axis = +75



- If lead I and aVF - positive
 - **Normal axis**
- If lead I - positive and aVF - negative
 - **Left axis deviation/normal**
- If lead I - negative and aVF - positive
 - **Right axis deviation**



Frontal plane axis

Rate 154 . Age not entered assumed to be 50 years old for purpose of ECG interpretation
 RR 390 . Sinus tachycardia.....rate> 99
 PR . Right bundle branch block.....QRSd>120, terminal axis(90,270)
 QRSd 118 . Minimal ST elevation, diffuse leads.....ST >0.10mV, ant/lat/inf
 QT 336 . Artifact in lead(s) V5 and baseline wander in lead(s) V5
 QTcB 538
 QTcF 460

Frontal plane axis = undetermined axis

--AXIS--

P 0

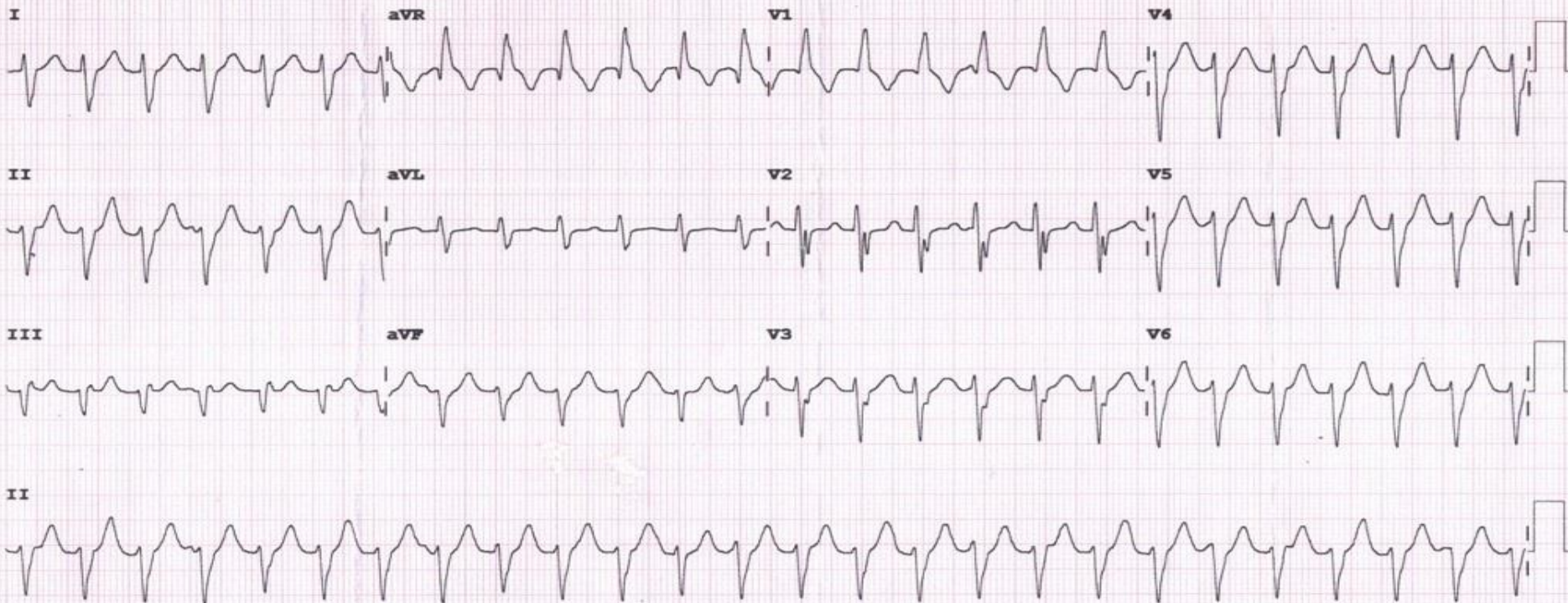
QRS -120

T 54

12 Lead; Standard Placement

- ABNORMAL ECG -

Unconfirmed Diagnosis



Device: Speed: 25 mm/sec Limb: 10 mm/mV Chest: 10.0 mm/mV

F 50~ 0.50~ 40 Hz W PH100B CL

ECG

interpretation

- Action potential
- Vector
- ECG waves & intervals
- Surface ECG leads
- Einthoven's triangle
- Special leads

CAD

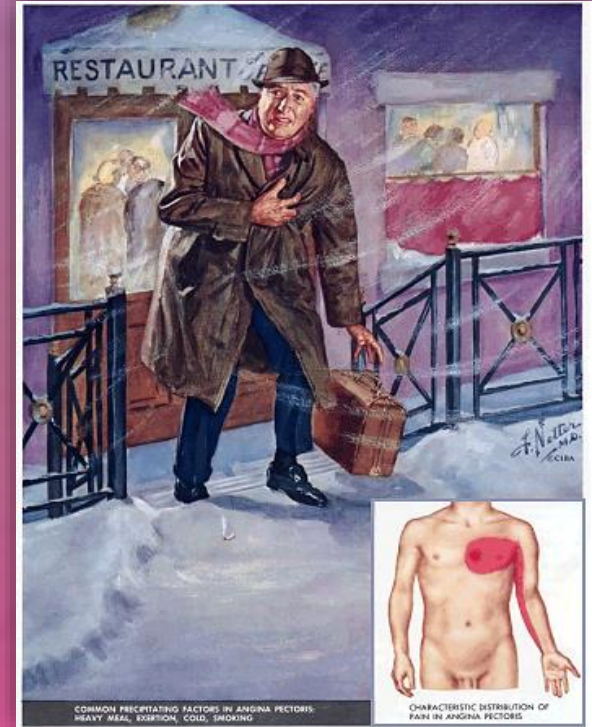
ECG

- Spectrum of CAD
- Pathophysiology
- Pattern of ischemic injury
- Clinical presentation & ECG correlation

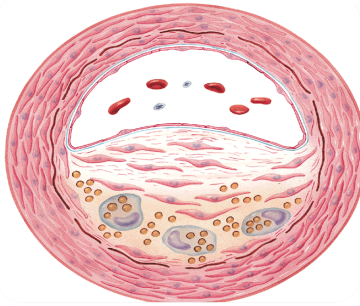
Coronary artery disease

● Coronary artery disease

- Spectrum of CAD
- Pathophysiology
- Pattern of ischemic injury
- Clinical presentation & ECG correlation



SPECTRUM OF CAD

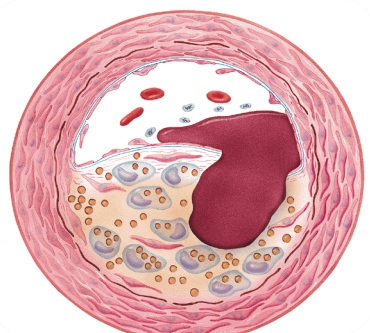


Chronic coronary syndrome (CCS)
Stable ischemic heart disease (SIHD)

Knuuti J, et al. Eur Heart J 2020;41:407-477
Fihn SD, et al. J Am Coll Cardiol 2014;64:1929-49



**Fixed obstruction of coronary artery by
atherosclerotic plaque**



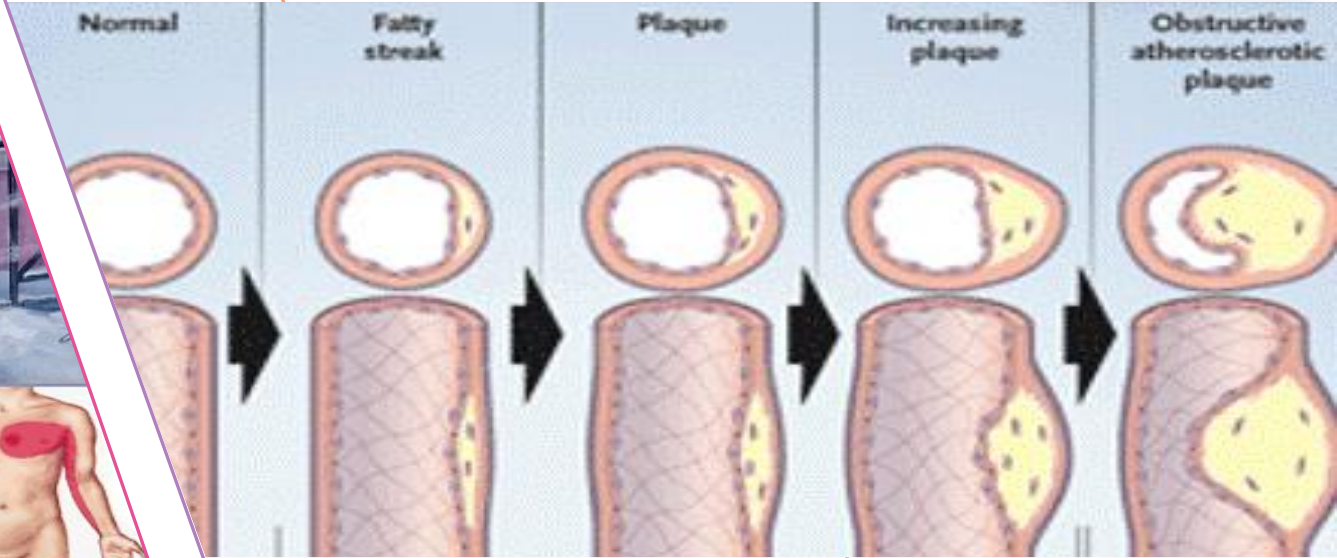
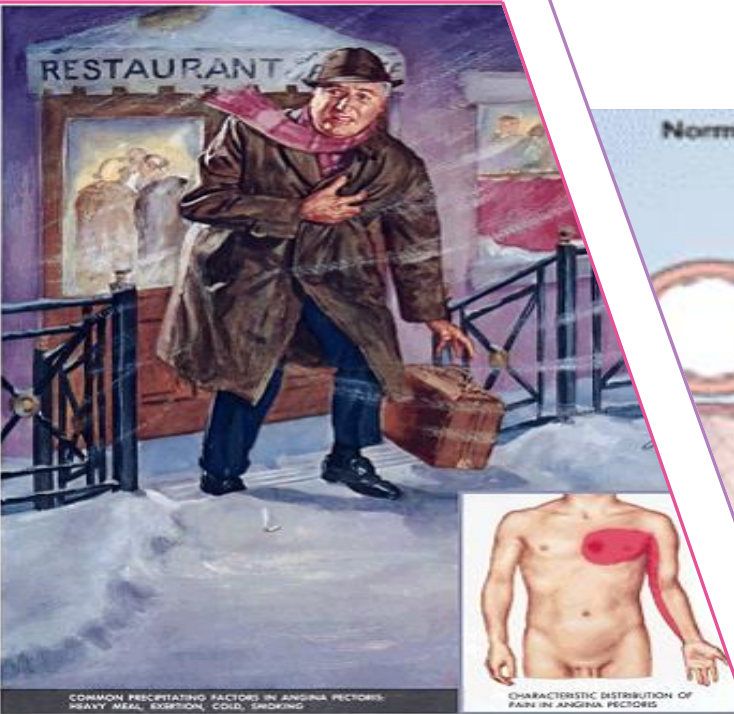
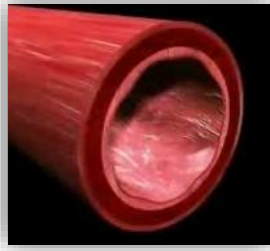
Acute coronary syndrome (ACS)

Ibanez B, et al. Eur Heart J 2018;39:119-177
Amsterdam EA, et al. J Am Coll Cardiol 2014;64:e139-e228



**Ruptured atherosclerotic plaque caused acute
coronary occlusion or stenosis**

Stable Ischemic Heart Disease

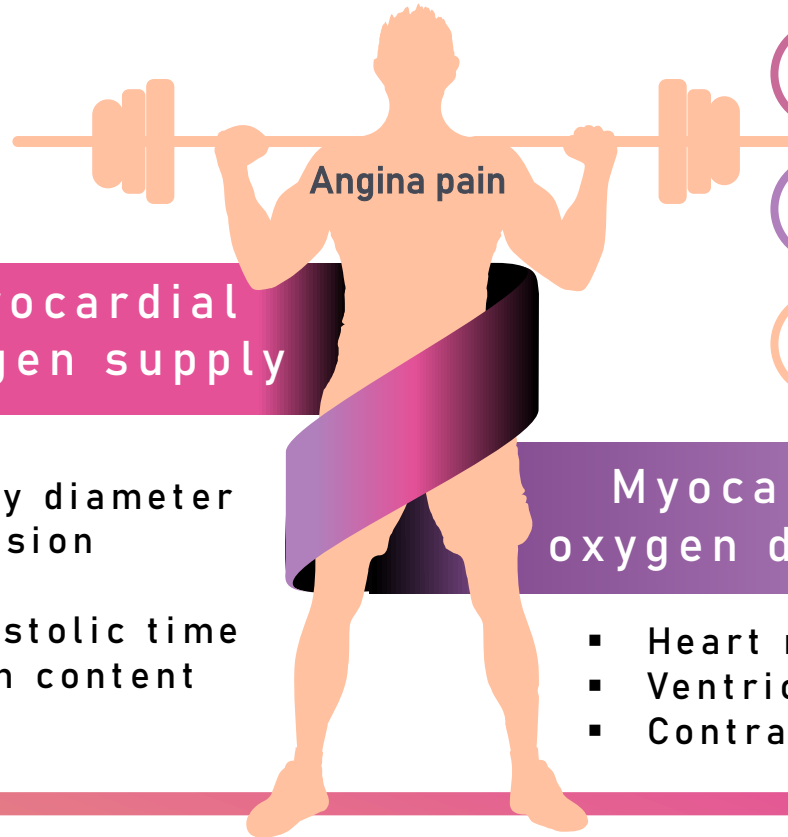


Noninvasive tests:
normal

Noninvasive tests:
abnormal

CHRONIC CORONARY SYNDROME

Myocardial ischemia



Constricting discomfort in the front of the chest or in the neck, jaw, shoulder, or arm



Precipitated by physical exertion



Relieved by rest or nitrates within 5 min

Myocardial
oxygen supply

Myocardial
oxygen demand

- Coronary artery diameter
- Coronary perfusion pressure
- Heart rate: diastolic time
- Arterial oxygen content

- Heart rate
- Ventricular wall tension
- Contractility



ANGINA SEVERITY



Angina only
with
strenuous
exertion

I



Angina with
moderate
exertion

II



Angina with
mild
exertion

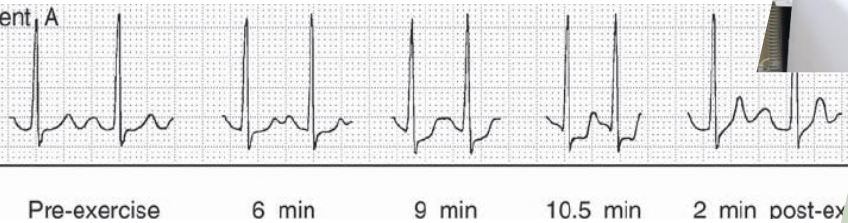
III



Angina at
rest

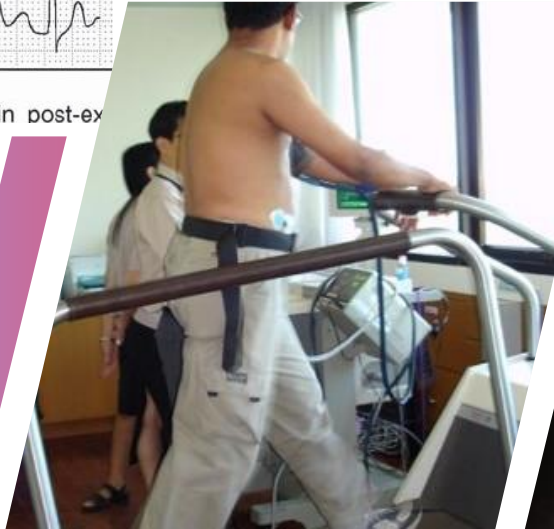
IV

Exercise Stress Test



Diagnosis

Risk
stratification



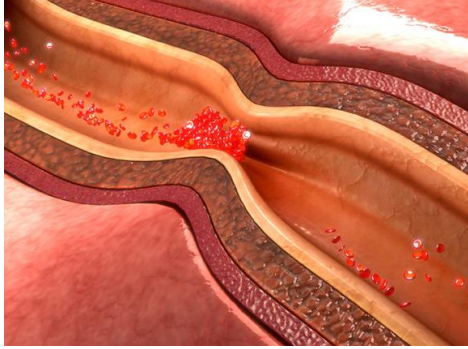
Supply ischemia, no infarct

Subendocardial infarct

Transmurular infarct

Occlusive & non-occlusive thrombus

Type 2 myocardial infarction



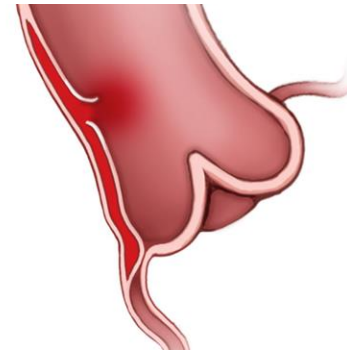
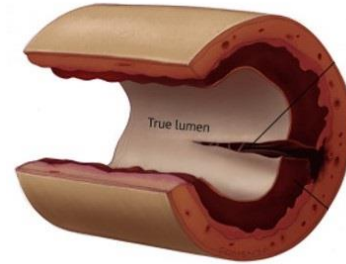
Coronary
Spasm
Prinzmetal angina



Coronary Embolism



Coronary
Dissection



Ascending
Aortic Root
Dissection

ACUTE CORONARY SYNDROME

2 UNSTABLE ANGINA

The plaque ruptures and a thrombus forms around the ruptured plaque, causing partial occlusion of the vessel. Angina pain occurs at rest or progresses rapidly over a short period of time.



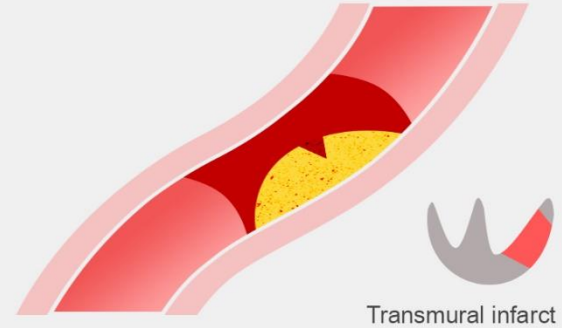
3 NSTEMI

During an NSTEMI, the plaque rupture and thrombus formation causes partial occlusion to the vessel that results in injury and infarct to the subendocardial myocardium.



4 STEMI

A STEMI is characterized by complete occlusion of the blood vessel lumen, resulting in transmural injury and infarct to the myocardium, which is reflected by ECG changes and a rise in troponins.



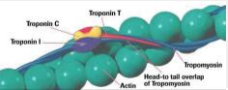
Normal, Inverted T waves, or ST depression



Normal, Inverted T waves, or ST depression



Hyperacute T waves or ST elevation



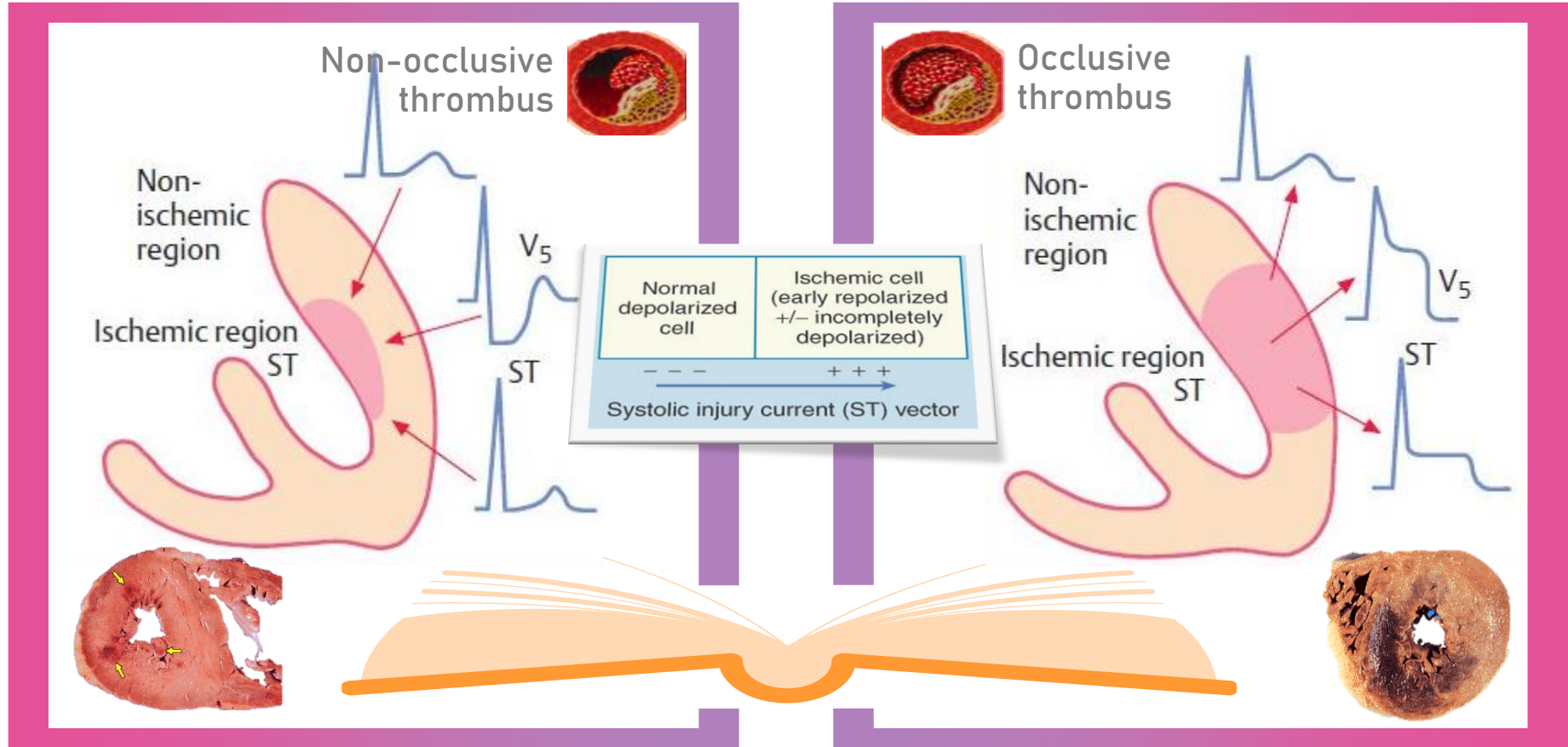
Normal

Elevated

Elevated

UA & NSTEMI

STEMI



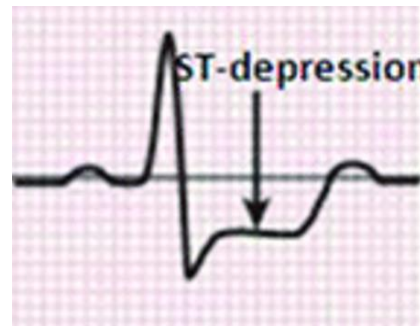
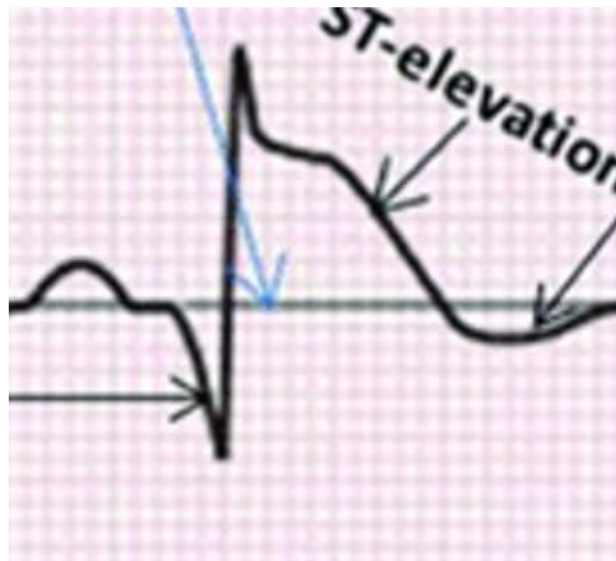
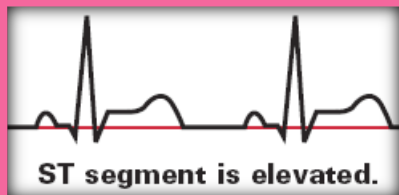
Subendocardial infarction

Transmural infarction

Universal definition

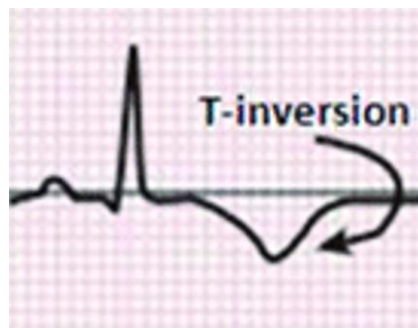
Myocardial Infarction

ST elevation
New ST elevation at
the J point in 2
contiguous leads with
cut-points...

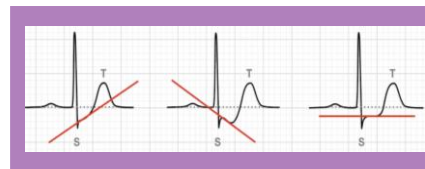


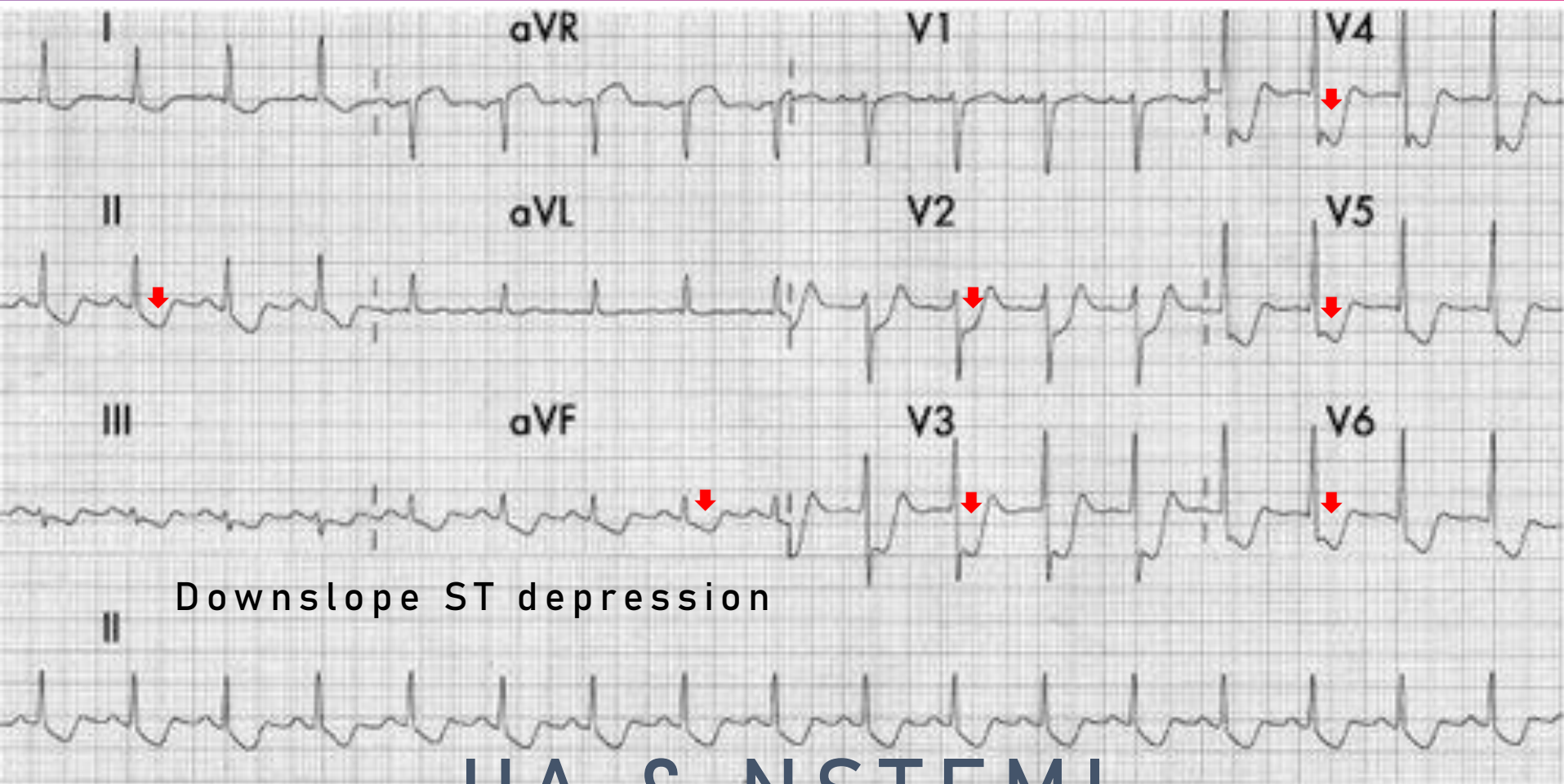
ST depression
New horizontal or
down sloping ST
depression ≥ 0.5
mm in 2 contiguous
leads

ST Elevation	<40 yo	>40 yo	All Ages
V2 -or- V3	>2.5 mm	>2 mm	>1.5 mm
ALL other Leads	>1 mm	>1 mm	>1 mm



T wave changes
T inversion > 1 mm in
2 contiguous leads
with prominent R
wave or R/S ratio > 1

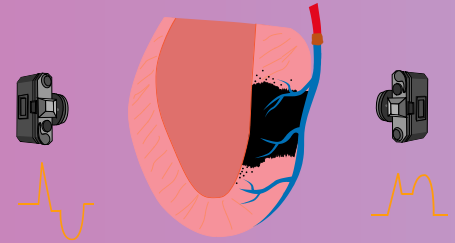


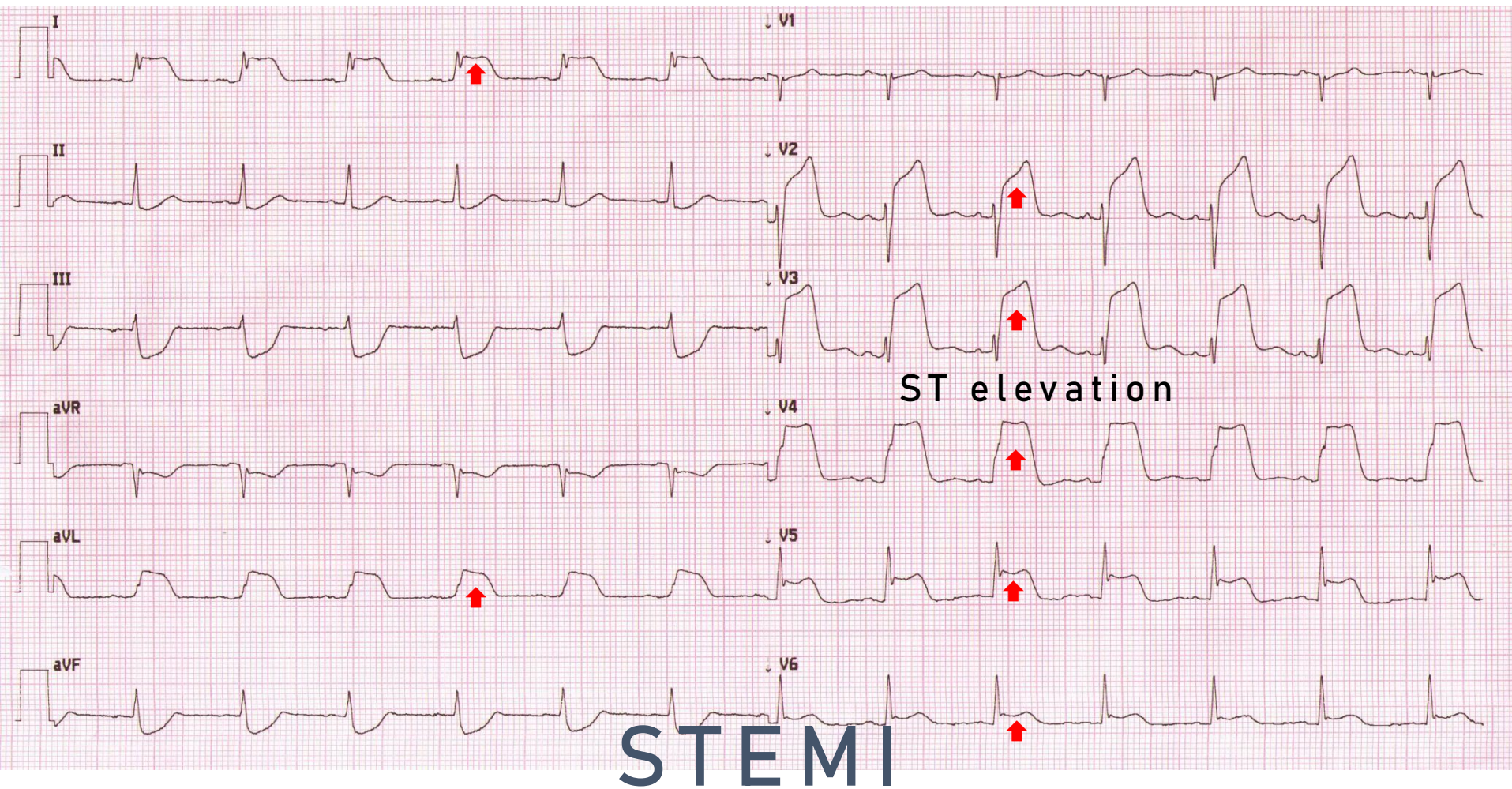


UA & NSTEMI

Causes of ST depression & T wave inversion

- ◆ Right or left bundle branch block (RBBB, LBBB)
- ◆ Right or left ventricular hypertrophy (LVH, RVH)
- ◆ Pre-excitation (WPW syndrome)
- ◆ Digitalis intoxication
- ◆ Hypokalemia
- ◆ Cerebral hemorrhage
- ◆ Reciprocal change of ST elevation

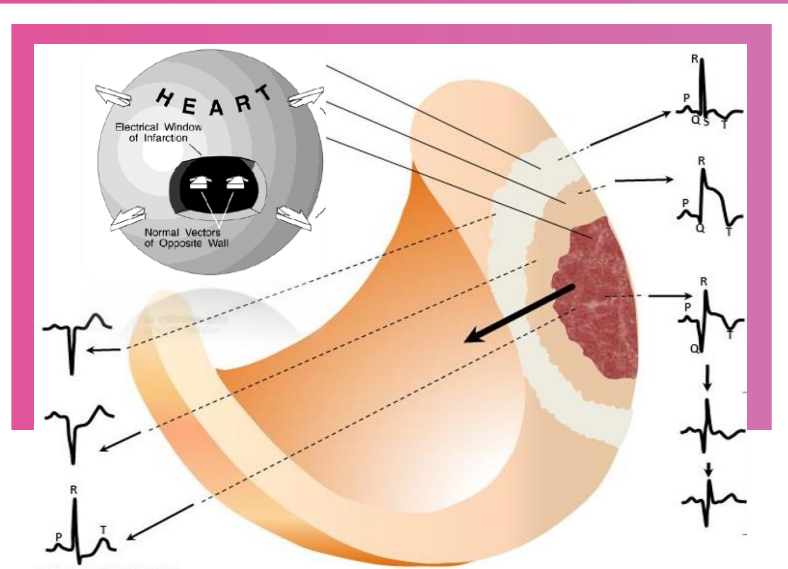




Causes of ST elevation

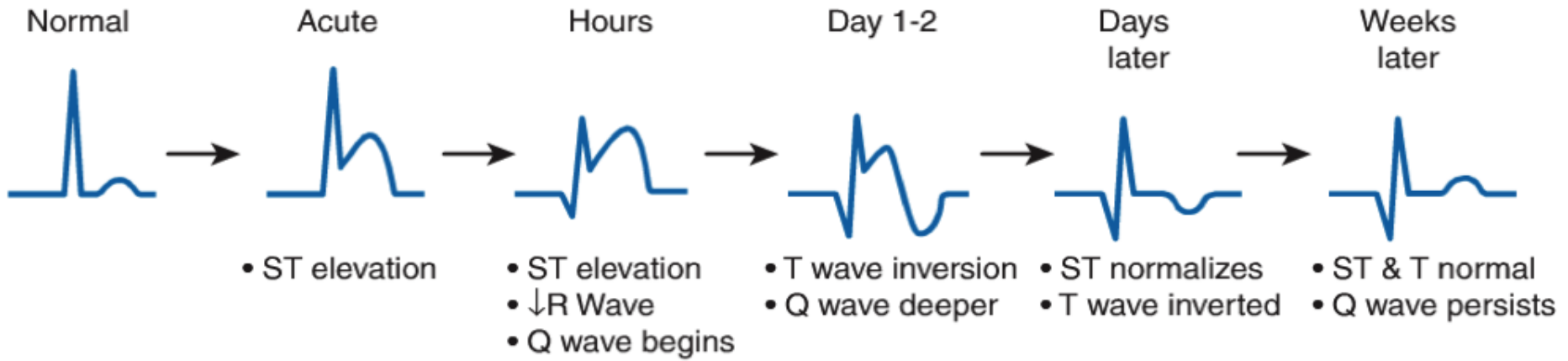
- ◆ Acute pericarditis
- ◆ LV hypertrophy
- ◆ Hyperkalemia
- ◆ Brugada syndrome
- ◆ Early repolarization syndrome
- ◆ Stress induced cardiomyopathy (broken heart syndrome/Takotsubo cardiomyopathy)



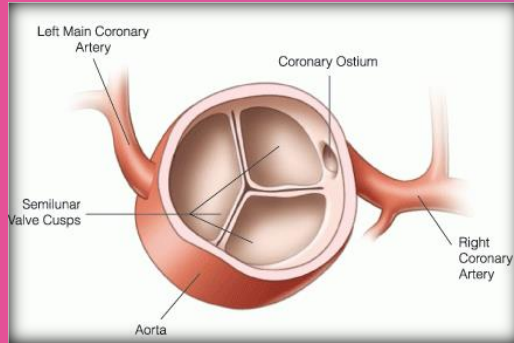


I lateral	aVR	V ₁ septal	V ₄ anterior
II inferior	aVL lateral	V ₂ septal	V ₅ lateral
III inferior	aVF inferior	V ₃ anterior	V ₆ lateral

EVOLUTION OF STEMI



Coronary artery

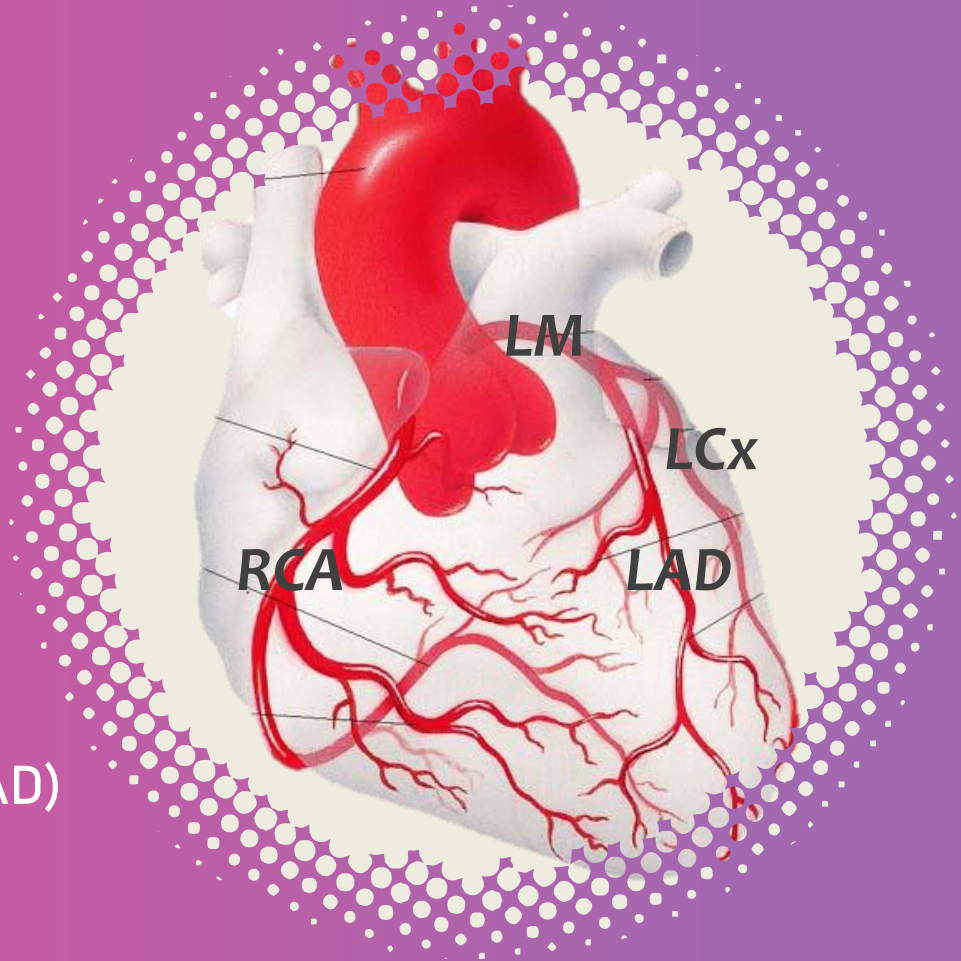


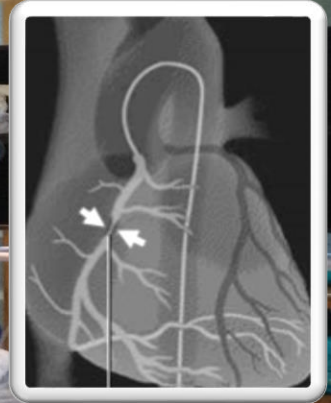
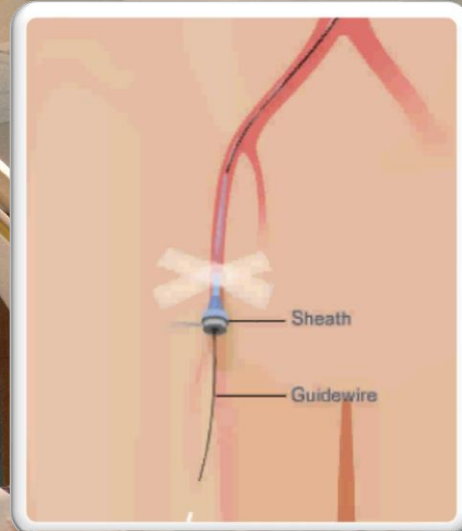
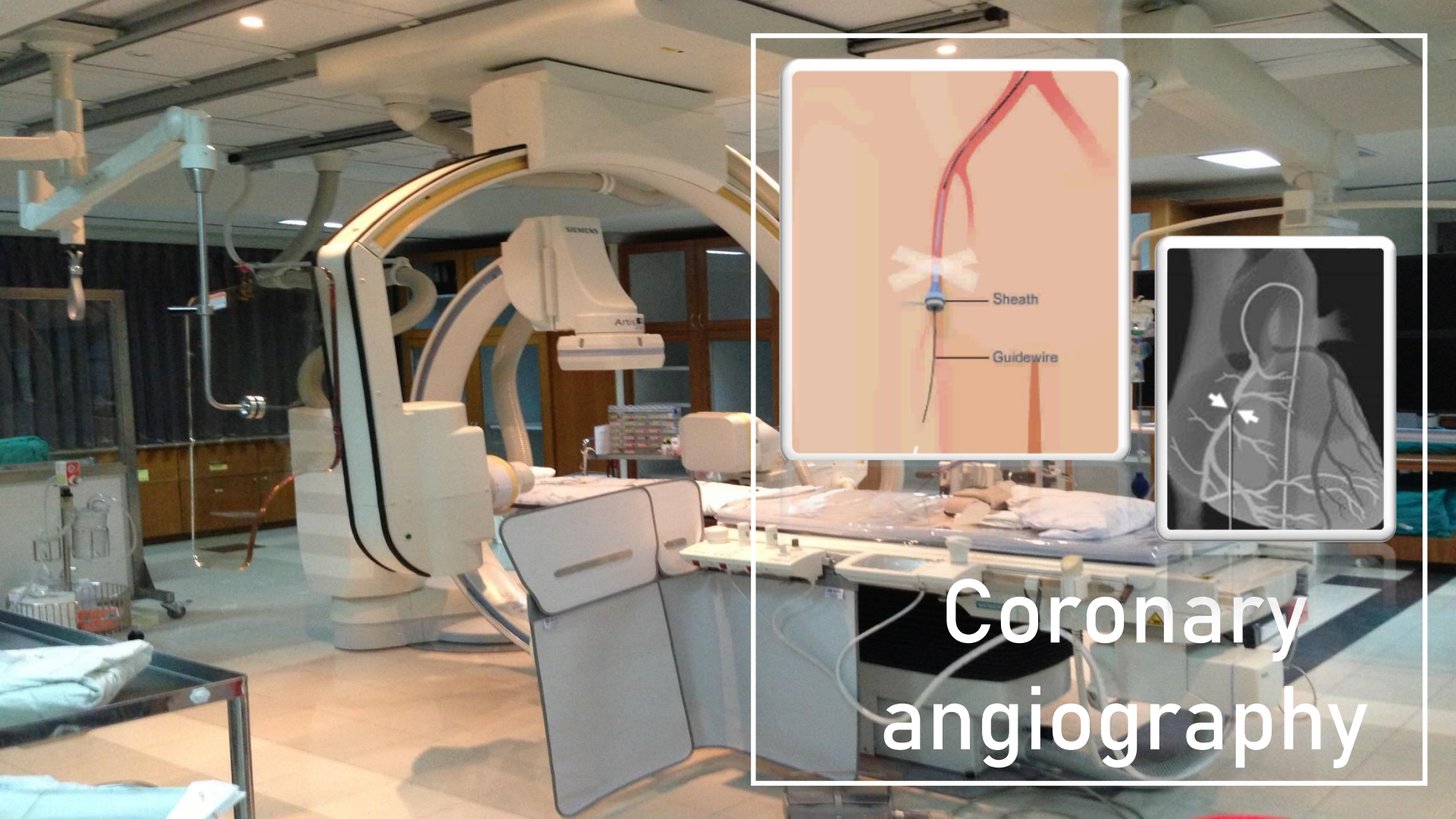
Left main coronary artery (LM)

Left anterior descending artery (LAD)

Left circumflex artery (LCx)

Right coronary artery (RCA)





Coronary
angiography

1

Right coronary artery (RCA)

- ◇ SA Node (60%)
- ◇ Right ventricle
- ◇ AV Node (>80%)
- ◇ Inferior wall of LV
- ◇ Posterior wall of LV (75%)

3

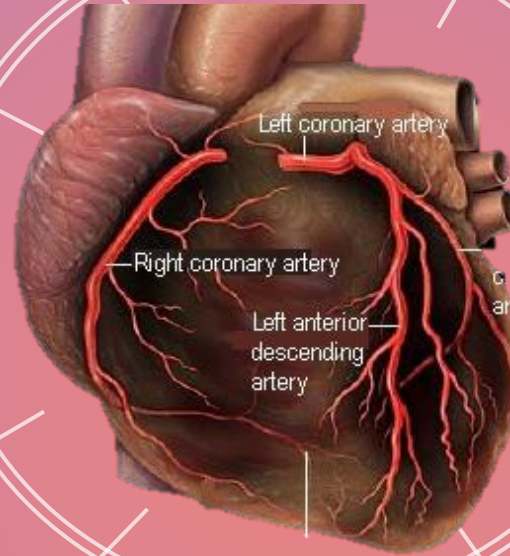
Left circumflex artery (LCX)

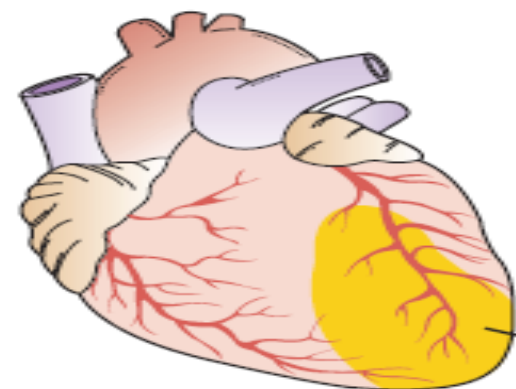
- ◇ SA node 40%
- ◇ AV node 20%
- ◇ Inferior, posterior and some part of lateral wall

2

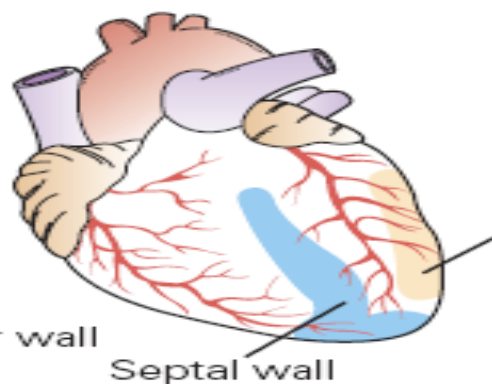
Left anterior descending artery (LAD)

- ◇ Septal wall, Anterior wall, Lateral wall
- ◇ Right bundle branch
- ◇ Left anterior fascicle

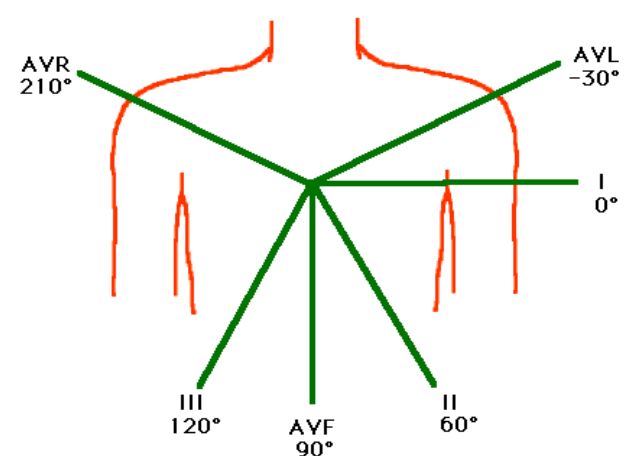




Anterior view

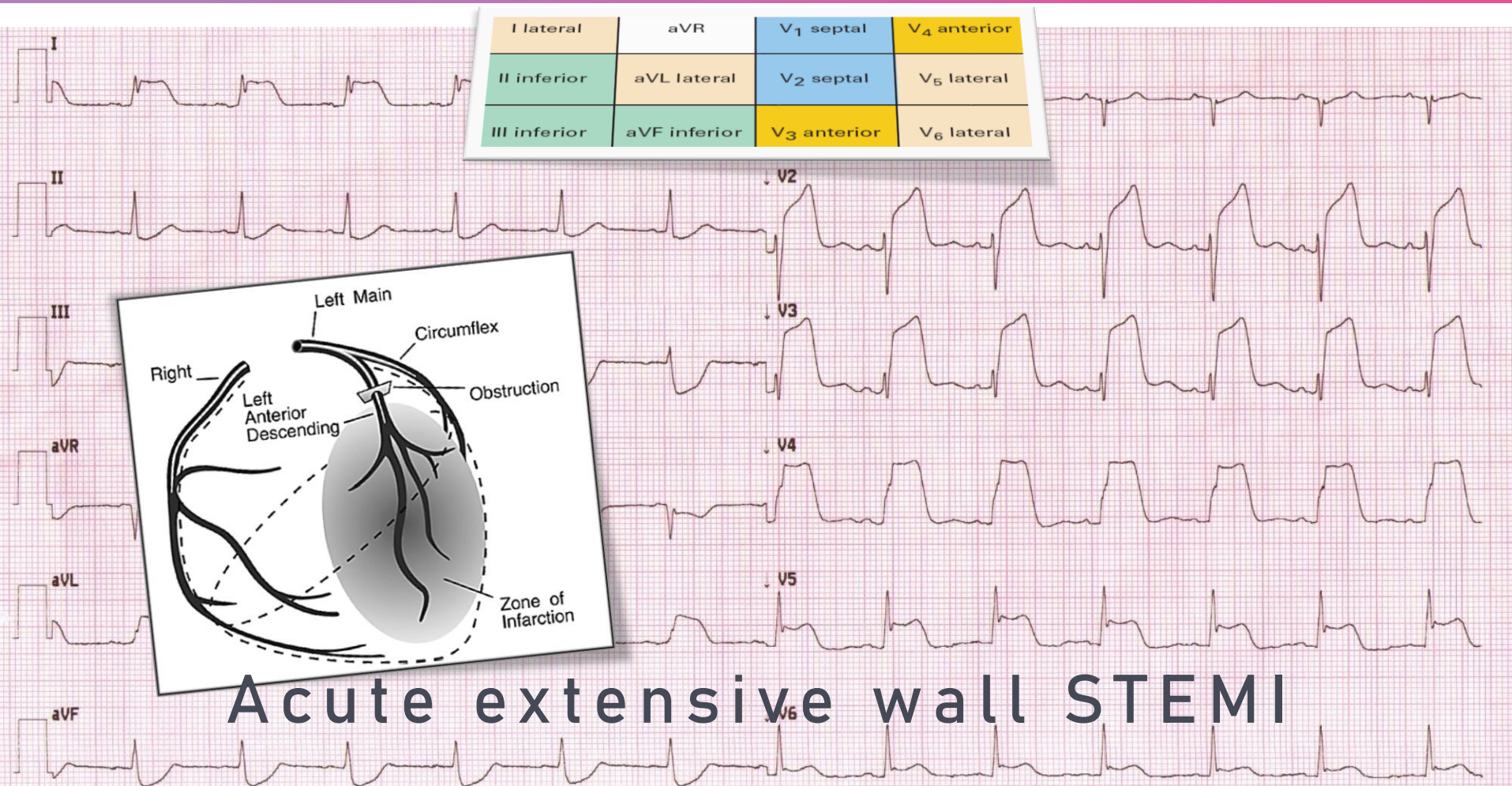


Anterior view



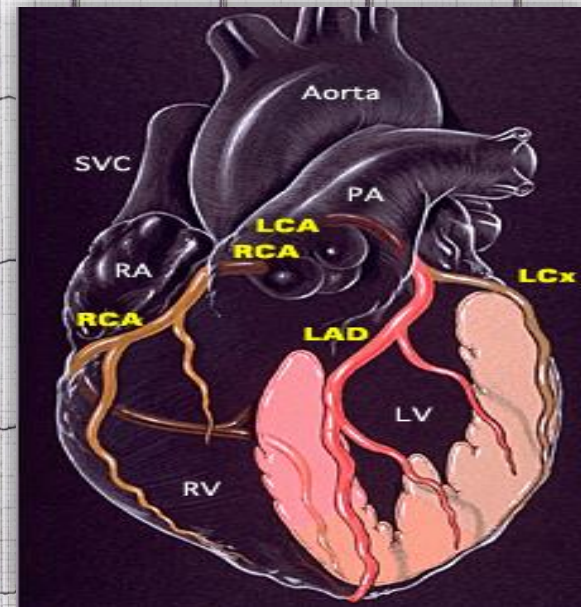
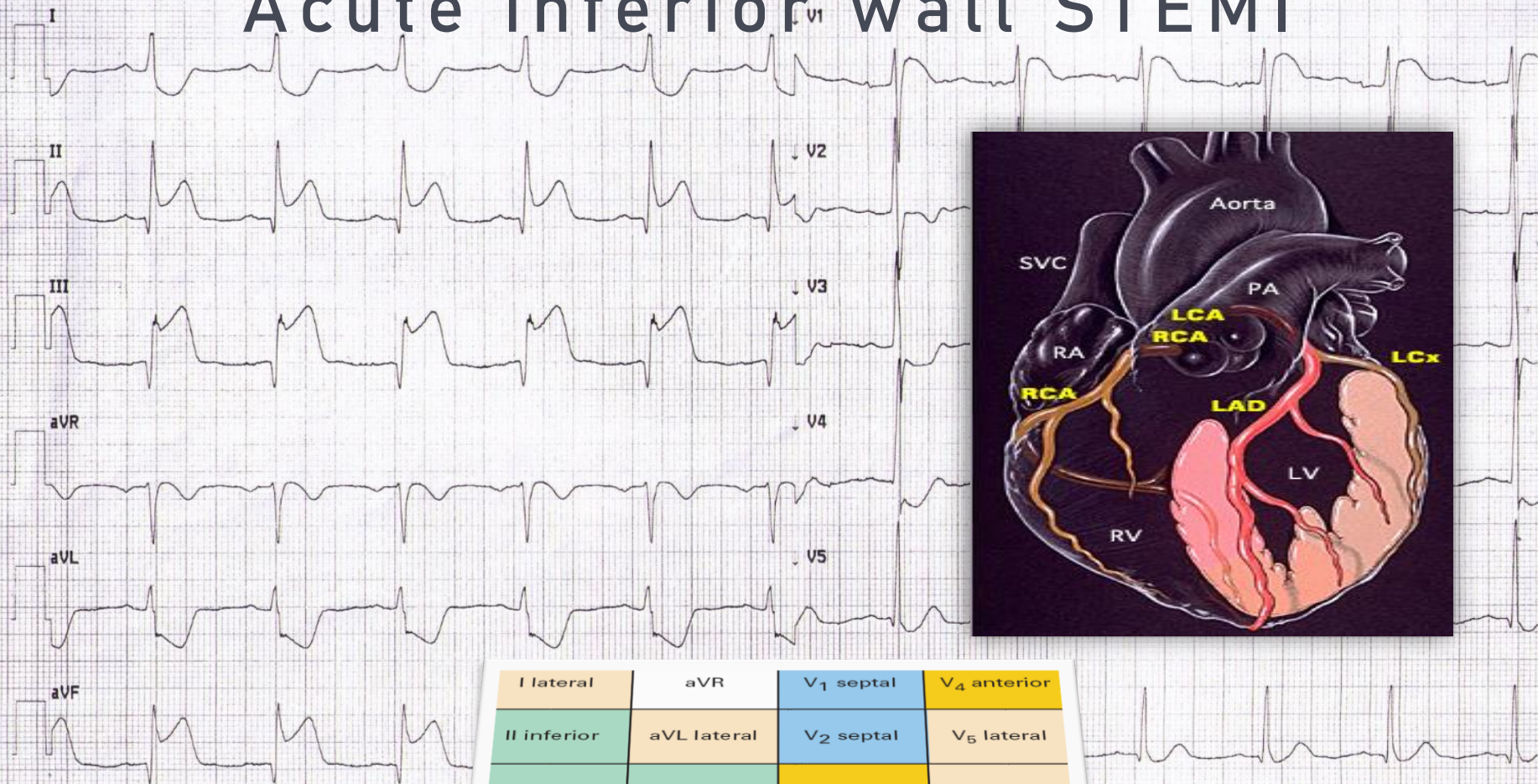
Location of MI by ECG Leads

I lateral	aVR	V ₁ septal	V ₄ anterior
II inferior	aVL lateral	V ₂ septal	V ₅ lateral
III inferior	aVF inferior	V ₃ anterior	V ₆ lateral

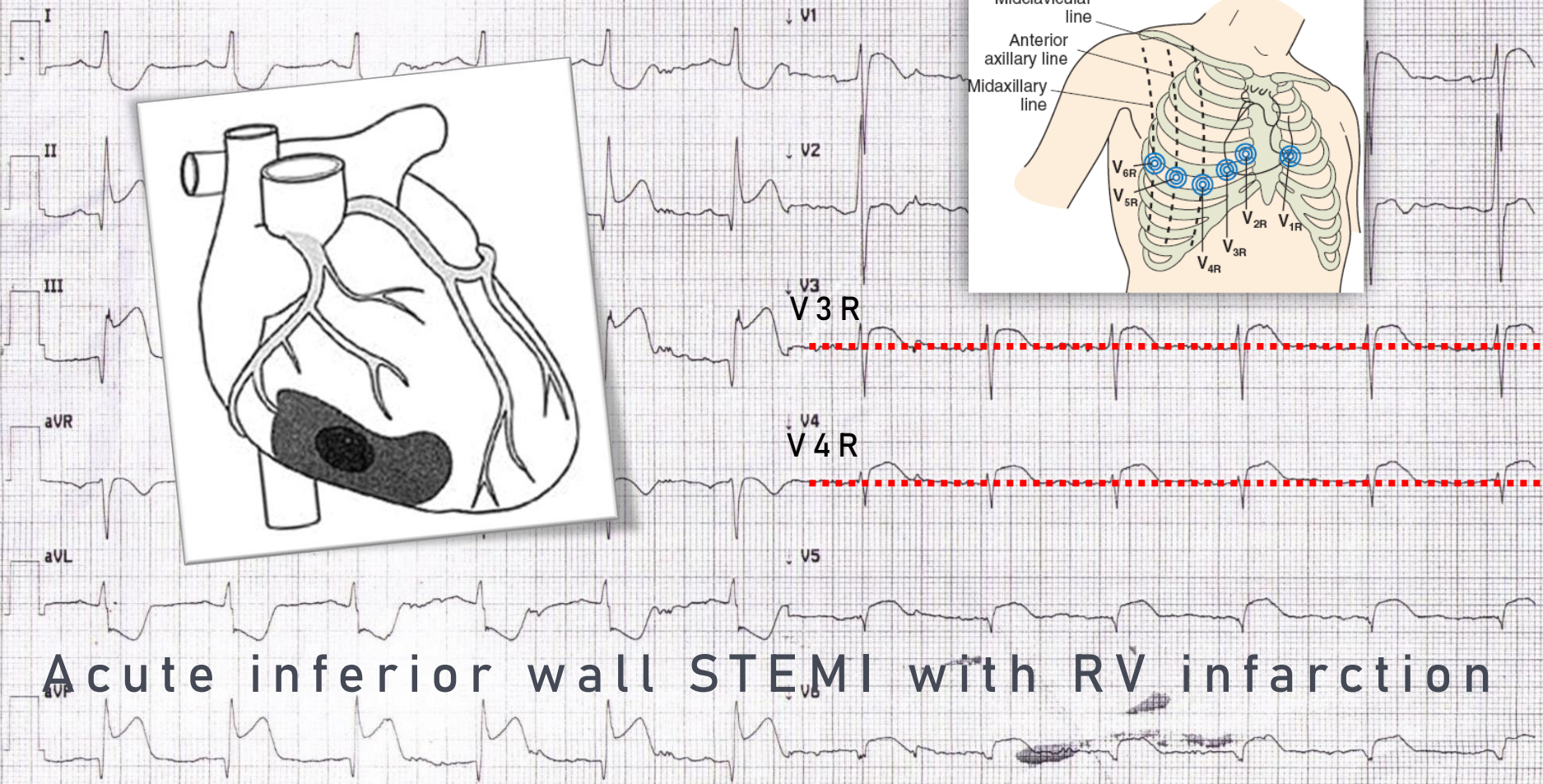


Acute extensive wall STEMI

Acute inferior wall STEMI

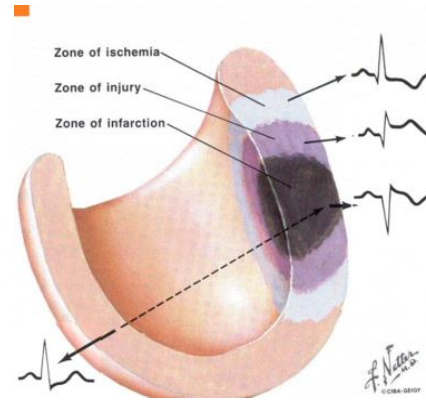
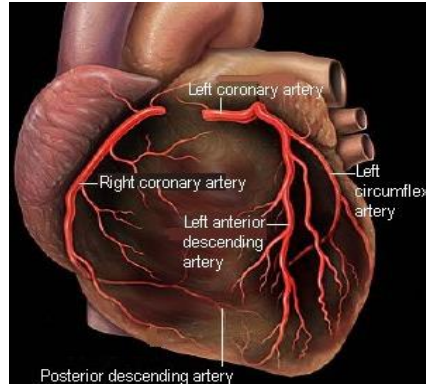
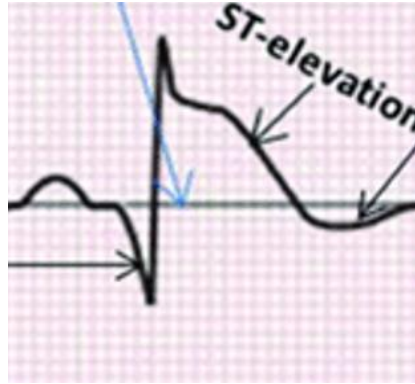


I lateral	aVR	V ₁ septal	V ₄ anterior
II inferior	aVL lateral	V ₂ septal	V ₅ lateral
III inferior	aVF inferior	V ₃ anterior	V ₆ lateral



ECG in clinical practice

Coronary artery disease



Ventricular Fibrillation



Diagnosis

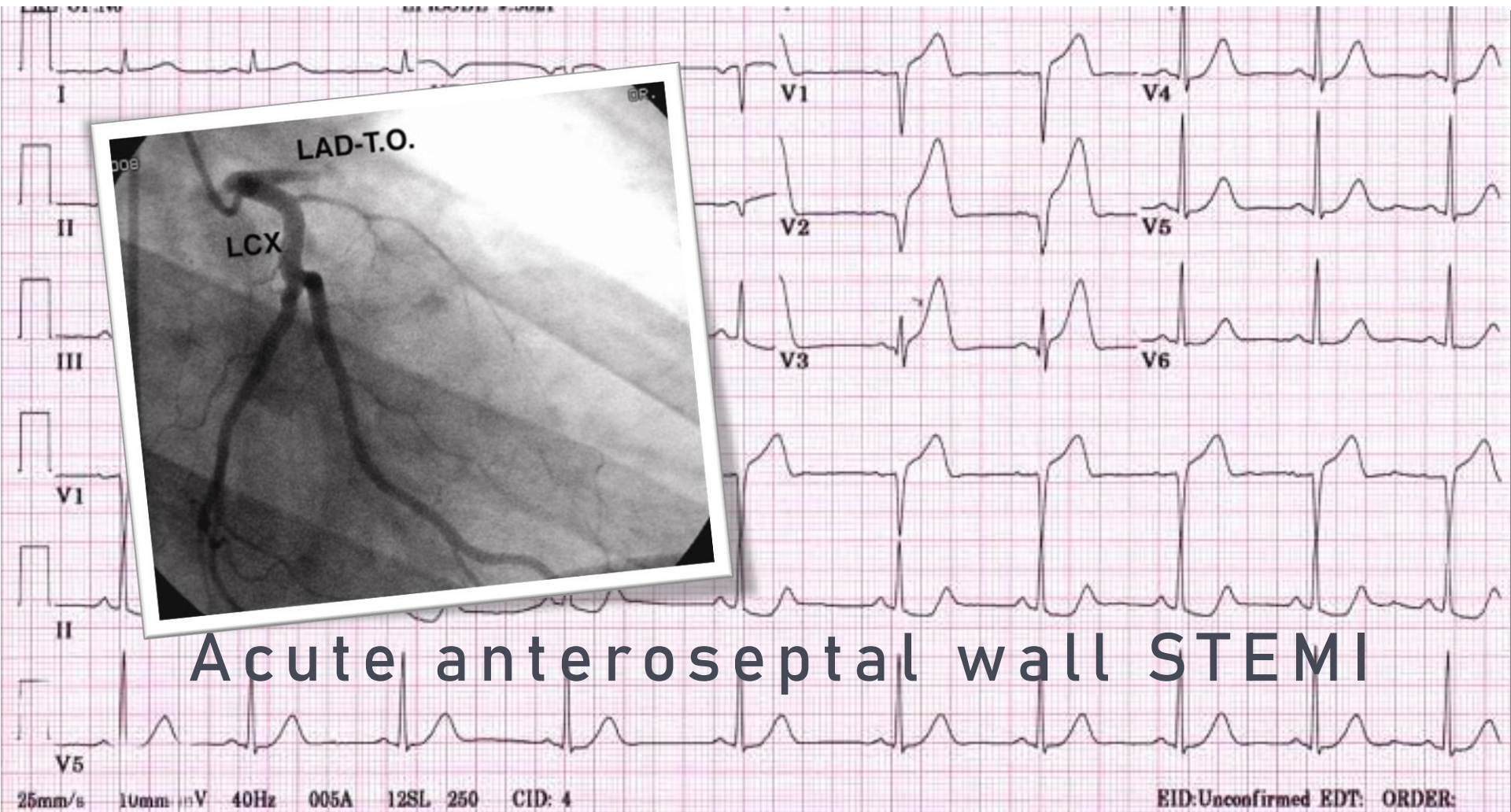
Prognosis

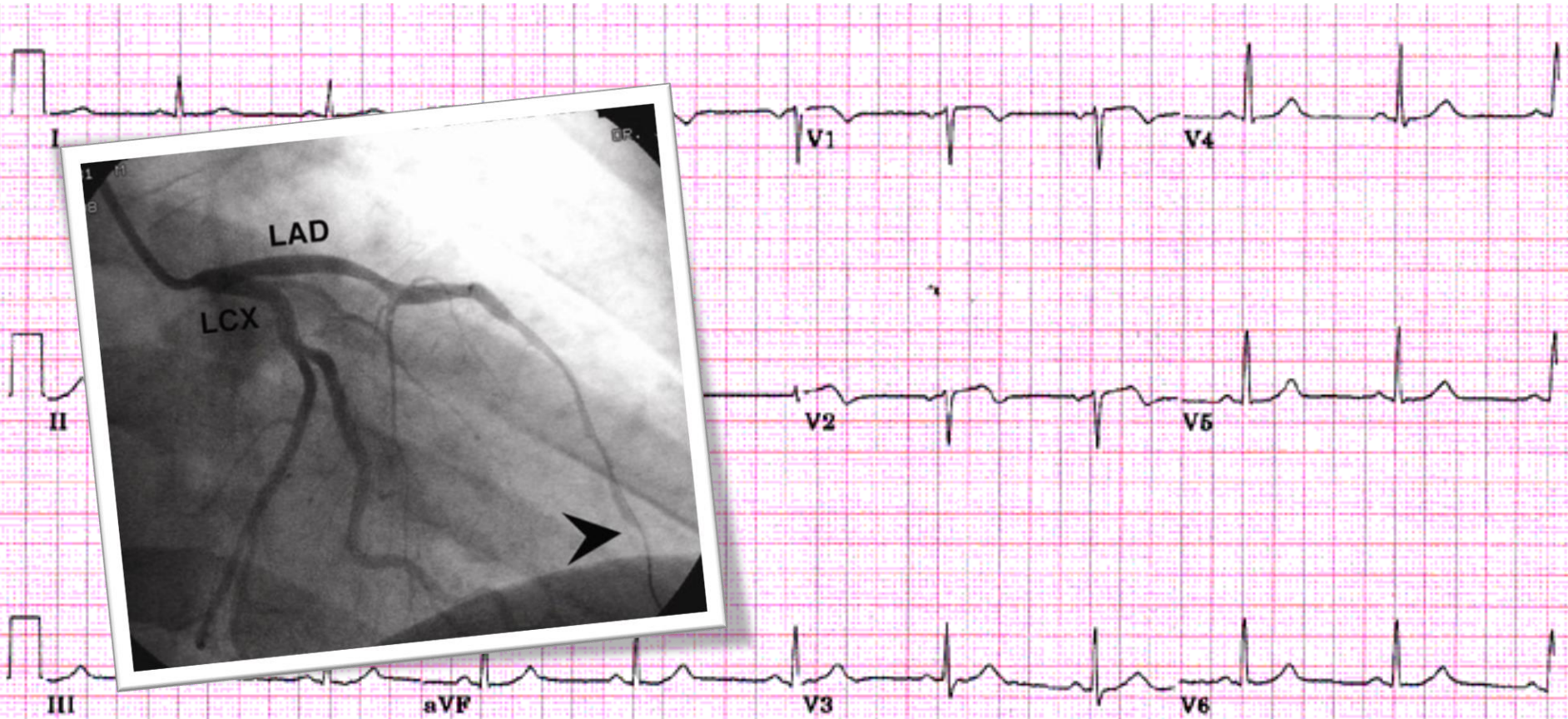
Monitor
treatment

Complication



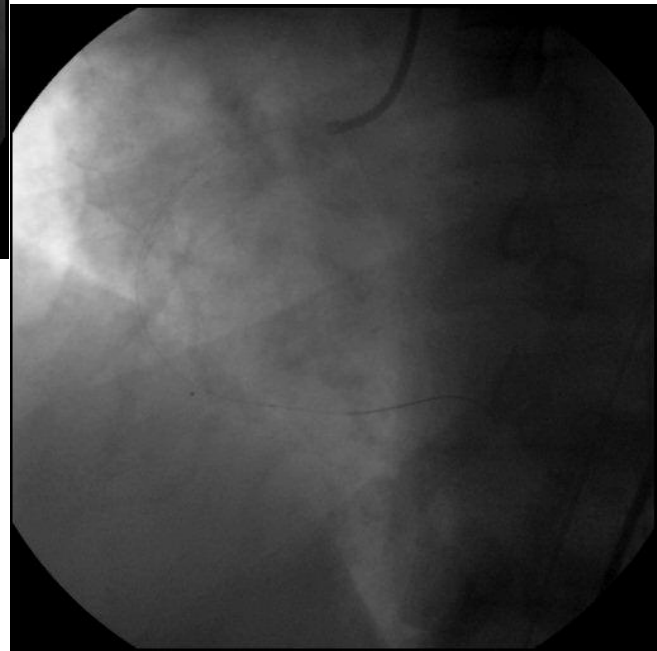
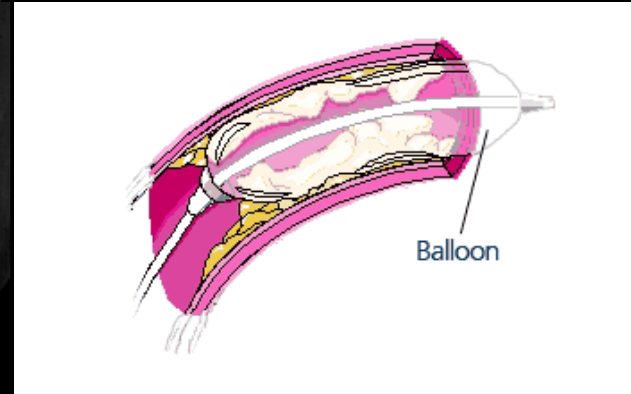
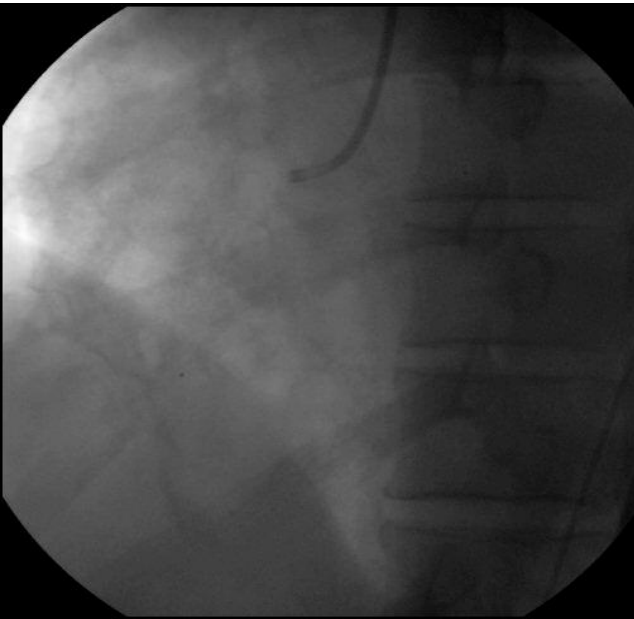
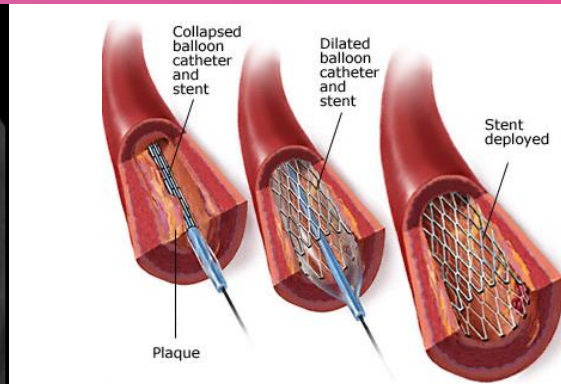
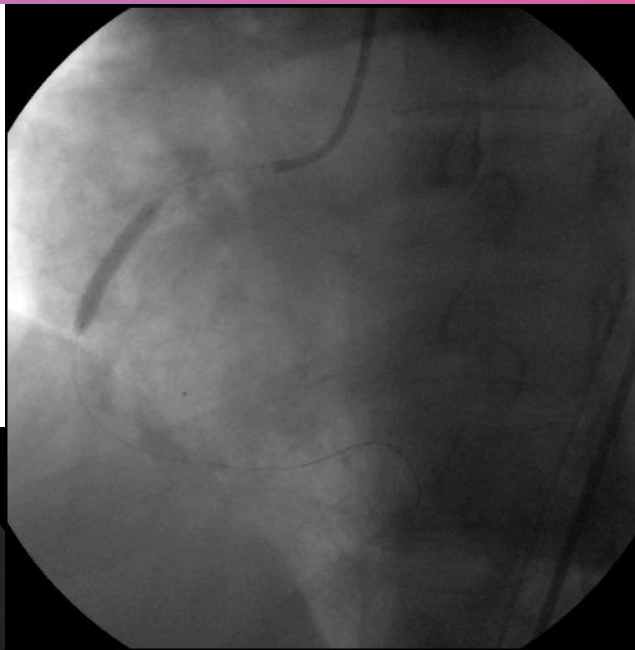
Acute anteroseptal wall STEMI



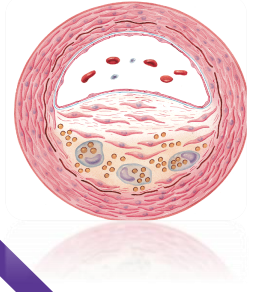


Acute anteroseptal wall STEMI

Percutaneous coronary intervention (PCI)



Chronic Coronary Syndrome

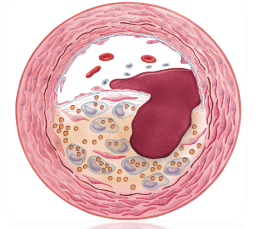
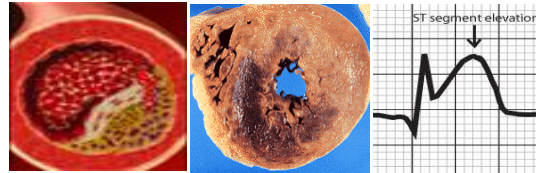
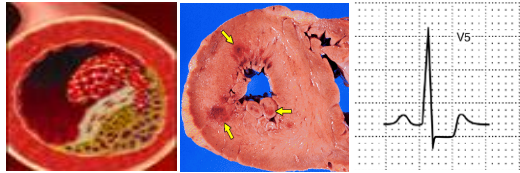


ANGINA PAIN



UA/NSTEMI

STEMI



Acute Coronary Syndrome

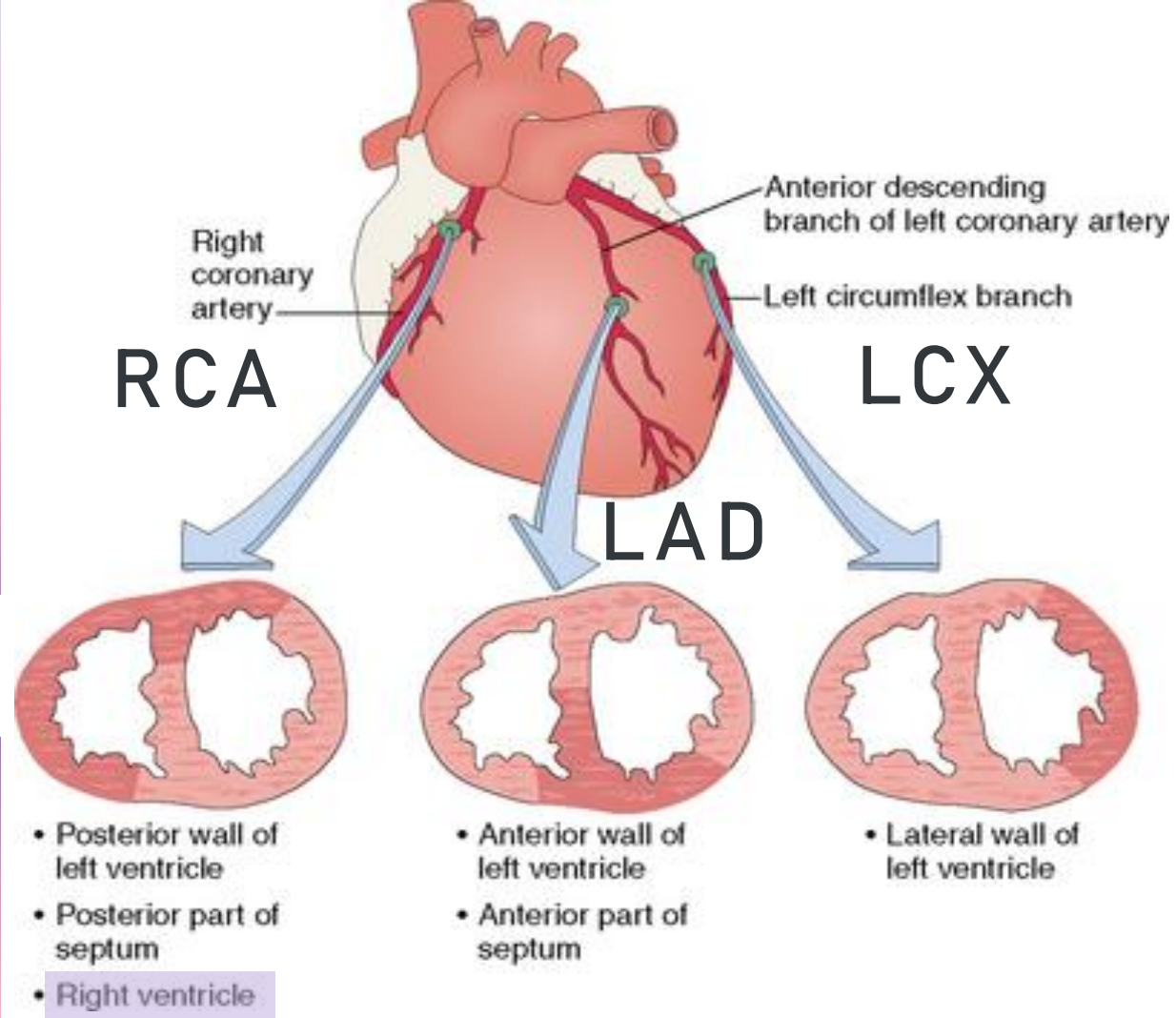
Location of MI by ECG Leads

I lateral	aVR	V ₁ septal	V ₄ anterior
II inferior	aVL lateral	V ₂ septal	V ₅ lateral
III inferior	aVF inferior	V ₃ anterior	V ₆ lateral

STEMI ECG

&

CORONARY
ARTERY





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mail: wtreechada@gmail.com